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Date: 1st June 2026

I hereby declare that, except where specifically indicated, the work submitted herein is my own original work.

Signed Hana Iza Kim ..... Date 1/6/26 .....

Modelling plasticity in the mouse olfactory bulb

Technical abstract

HANA IZA KIM - Pembroke College, 1st June 2026
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From the chemistry of a single cell to the workings of complex organisms, life depends on homeostasis, the ability to keep internal conditions within narrow, precise boundaries despite external fluctuations. In the nervous system, this is achieved through homeostatic plasticity (HP), which adjusts synaptic strength and intrinsic cell excitability to maintain activity within a functional range. Although HP is observed widely across species, its underlying mechanisms remain poorly understood, partly because the interconnected nature of neural circuits and their inherently difficult accessibility. Computational modelling offers a powerful complementary approach, and in this work we develop a biologically inspired computational model of the mouse olfactory bulb (OB) to identify parameter changes that preserve circuit output consistent with experimentally observed HP changes.

A minimal model of the OB is shown in Fig. 1, which incorporates the key excitatory and inhibitory cell populations: anaxonic (blue) interneurons, axonic (green) interneurons, external tufted cells (ETCs, red), mitral cells, and granule cells. The circuit includes two coupled glomeruli, the minimal architecture needed to capture inter-glomerular lateral inhibition. Granule cells are excluded due to poorly constrained cortical feedback and steady-state firing rates are modelled using the Naka-Rushton function with cell-type-specific Hill exponents fitted to *in vitro* data, initially all assumed to be fixed. A twelve-dimensional parameter vector θ encodes all synaptic weights. Baseline configurations are identified by minimising a least-squares loss targeting normalised firing rates of 0.5 at unit input using the Adam optimiser over 1000 random initialisations. Of these, 570 converge to valid solutions, all of which are confirmed to be locally stable through linearisation of the network dynamics.

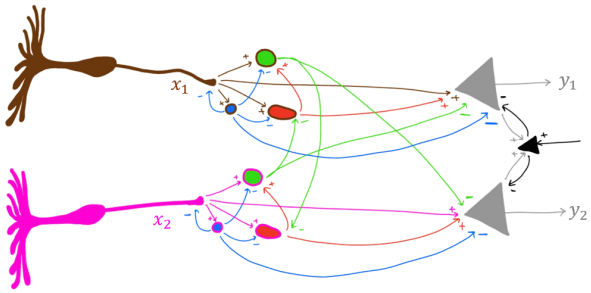


Figure 1: Simplified OB circuit used in the model. Excitatory (+) and inhibitory (-) connections between the blue (b), red (r), green (g), mitral (y), and granule (gr) populations.

To characterise output-preserving parameter changes, the mitral cell output y is expanded in a second-order Taylor series about each baseline θ_o:

$$y(\theta_o + \Delta\theta) \approx y(\theta_o) + \text{grad}^T \Delta\theta + \frac{1}{2}\Delta\theta^T \mathbf{H}\, \Delta\theta + \mathcal{O}(\|\Delta\theta\|^3) \quad (1)$$

where grad $= \nabla_\theta y(\theta_o)$ is the output gradient with respect to the parameters, and $\mathbf{H} = \nabla^2_\theta y(\theta_o)$ is the corresponding Hessian matrix. The output-preserving directions $\Delta\theta$ must therefore be orthogonal to the gradient ($\text{grad}^T \Delta\theta = 0$) and aligned with eigenvectors of the Hessian with near-zero eigenvalues.

Across all 570 baselines, the Hessian consistently exhibits four near-zero eigenvalues, revealing a robust four-dimensional output-preserving subspace. Singular value decomposition of the stacked eigenvectors confirms that four dominant directions recur consistently across parameter configurations. A sparse rotation, minimising the L_1 norm of the basis under an orthogonality constraint, produces interpretable basis vectors. Sub-

sequent iterative signed-permutation grouping identifies four distinct sparse basis families. Stepping along these directions in parameter space confirms that mitral output changes by less than 1% for perturbations of 100% relative to $\|\theta\|$.

One of the basis vectors contains a dominant $w_{b,g}$ component, consistent with experimentally observed increase in inhibitory drive from blue to green cells. The vector also specifies an increase in $w_{\mathrm{osn},g}$ and $w_{r,g}$ as compensatory adjustments required to preserve mitral output. Biologically, these represent stronger excitatory drive from OSNs and ETCs onto the axonic cell population, which is plausible and consistent with classical homeostatic synaptic upscaling in response to reduced sensory input.

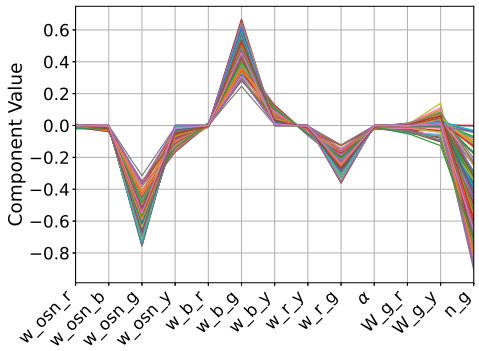


Figure 2: 187 candidate directions satisfying biologically observed decrease in n_g and increase in $w_{b,g}$.

However, these results are incomplete, as the Hill exponents n are assumed to be constant. To capture the experimentally observed axon initial segment (AIS) shrinkage, n_g is incorporated as a free parameter in the following analysis. Because n_g has no effect on circuit output at the standard operating point $x = 1$ under the normalised Naka–Rushton parameterisation, a second operating point ($x = 0.5$, $y_{target} = 0.2$) is used to introduce sensitivity. Of 1,000 initialisations, 214 satisfy both constraints and are confirmed to be stable. A constrained direction analysis pins a unit step in n_g and solves to find the compensatory weight adjustments that minimise the resulting output change via the pseudoinverse of a partitioned Hessian. Of the 214 candidate directions, 187 satisfy the experimental constraints of decreasing n_g and increasing $w_{b,g}$ (Fig. 2).

Their average provides a single representative compensatory direction, which predicts coordinated suppression of green cell activity through reductions in excitatory $w_{\mathrm{osn},g}$ and $w_{r,g}$ alongside the increase in inhibitory $w_{b,g}$ and decrease in n_g. To preserve mitral output under reduced green cell inhibition, the model further predicts an increase in $w_{b,y}$ and a decrease in $w_{\mathrm{osn},y}$. Stepping along this direction produces substantially smaller output changes than random perturbations, though larger than the null-space directions (Fig. 3). While these changes are consistent with existing experimental evidence and provide concrete targets for future experimental work, the conclusions remain not entirely satisfactory, as perturbations in these directions produce small but measurable changes in output y.

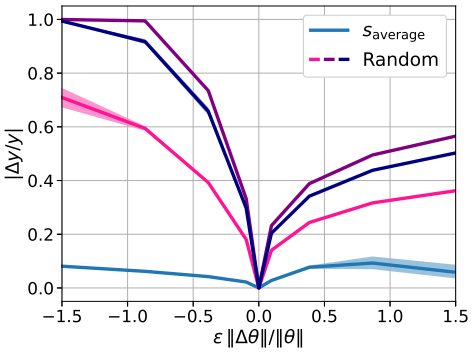


Figure 3: Average relative change in mitral cell output as a function of normalised step size $\varepsilon\|\Delta\theta\|/\|\theta_o\|$ along the average direction when $x = 1$. Shaded bands denote 95% CIs.

Several natural extensions follow from this work and could improve the biological realism of the model. These include allowing excitability changes in additional cell populations and explicitly modelling all Naka–Rushton parameters rather than only the Hill coefficient, incorporating activity-dependent changes in OSN signalling observed during sensory deprivation, and introducing granule cells as an explicit circuit component.

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1 Introduction

From the chemistry of a single cell to the workings of complex organisms, life depends on a remarkable ability for self-regulation. No matter how much the external environment fluctuates, living organisms must keep their internal conditions within narrow, precise boundaries. This process is known as homeostasis[1]. Common examples include the regulation of body temperature, blood glucose concentration, and pH balance. These variables are kept within physiological limits through feedback mechanisms that detect and correct any deviation from their set point.

Maintaining stability in biological systems is inherently challenging given that external perturbations are often stochastic, internal components can introduce noise, and delays in feedback can destabilise regulation. Neural circuits face similar challenges. Not only must they remain stable, but they must also stay flexible enough to support learning and adaptation. The consequences of failing to meet either are serious. Insufficient activity can impair information transmission and circuit responsiveness, while excessive activity can lead to disorders such as epilepsy or chronic pain.[2,3] It is this risk of under or over excitation that makes homeostatic regulation in the nervous system so essential.

To keep activity within a functional range, neurons rely on homeostatic plasticity (HP), which adjusts both synaptic strength and intrinsic excitability in response to prolonged changes in activity levels.[4] Synaptic scaling multiplies all incoming synaptic weights by a uniform factor, stabilising firing rates while preserving the relative differences that encode learned information.[5] Intrinsic plasticity complements this by modifying ion channel expression, tuning a neuron's firing threshold and gain. Together, these mechanisms regulate the broader balance between excitation and inhibition across the network[6].

HP is recognised as a widespread feature of the nervous system, observed across cortical and subcortical structures and across a range of species.[3,4,7,8] However, despite its importance, many aspects of the underlying mechanisms remain poorly understood. A central reason for this is the inherent difficulty of studying plasticity in living neural tissue. Neural circuits are highly interconnected, meaning that any manipulation targeting one component tends to propagate unpredictably across the network, making it difficult to isolate the contribution of a specific mechanism. Recording from neurons *in vivo* adds further technical demands, and the very act of intervening risks perturbing the system under investigation.

These challenges make computational modelling a powerful complementary approach. By design, models allow individual components and plasticity rules to be varied in isolation. Constructing biologically constrained models makes it possible to test hypotheses that are difficult to carry out experimentally and to generate predictions for future studies.

In this work, we use computational modelling to study homeostatic plasticity in the mouse olfactory bulb (OB). The OB is a particularly well-suited system, being compact and well-characterised with clearly defined cell types, and benefiting from detailed experimental measurements of activity dynamics and circuit structure that provide the biological constraints needed to build and validate a realistic model.[9] Our ultimate aim is to identify mechanisms underlying homeostasis in the OB, provide testable predictions, and guide future experimental work.

The remainder of this thesis is organised as follows. Section 2 introduces the olfactory bulb circuit, describing the key neuron types and the sensory deprivation experiment that provides the experimental motivation for this work. Section 3 presents the mathematical model developed, covering the neuronal response function, parameter fitting, fixed-point

solution, and stability analysis. Section 4 characterises the output-preserving subspace of the circuit, identifying sparse directions in parameter space along which synaptic weights can change without altering the network output. Section 5 incorporates the experimental constraints from sensory deprivation, extending the analysis to include changes in intrinsic excitability and identifying the compensatory adjustments required to maintain stable circuit output.

All simulations and analyses were implemented in Python. The full codebase, including simulation scripts and analysis notebooks, is publicly available at `https://github.com/hanaizakim/4th_yr_project.git`.[10]

2 Olfactory Bulb (OB)

2.1 OB: Sensory Deprivation

Sensory deprivation has been shown to induce homeostatic plasticity (HP) in the mouse olfactory bulb (OB),[8] and the procedure used to achieve this is both simple and minimally invasive. Because mice have two olfactory bulbs and input is ipsilateral,[8,11] blocking one nostril selectively reduces sensory drive to the bulb on the same side while leaving the bulb on the other side largely unaffected. This anatomical segregation makes the system particularly well-suited for controlled deprivation experiments. A temporary nose plug inserted into one nostril is sufficient to reduce sensory input reversibly, while allowing the animal to continue breathing normally through the other side, without permanent physiological damage.

Using this approach, experiments have shown that 24 hours of sensory deprivation induces measurable structural changes within the OB. In particular, dopaminergic axonic neurons exhibit a reduction in axon initial segment (AIS) length following the deprivation period.[8] This type of structural adaptation can be interpreted as a homeostatic response in which neural elements adjust their connectivity and intrinsic excitability in response to the level of incoming sensory stimulation.

Strikingly, when the nose plug is removed, OB output returns to its original activity level despite the structural changes induced during the sensory deprivation period.[12] Since dopaminergic axonic AIS length is reduced, the restoration of circuit output implies that compensatory changes must arise elsewhere in the network. Identifying the mechanisms underlying this compensation is the central question motivating the work presented here.

2.2 OB: Key Neuron Types

To investigate possible mechanisms of homeostatic plasticity, the first stage of the project involved developing a mathematical model of the olfactory bulb (OB) that captures its main neuronal populations and their interactions.[11] Rather than attempting to represent every cell type present, the model focuses on the core populations whose dynamics are most relevant to sensory processing and circuit stability (Figure 4). This kind of reduced, population-level approach is standard in computational neuroscience, where the goal is to capture dominant circuit behaviour rather than to account for every biological detail.[13]

The OB is the first central processing stage of olfactory information, transforming high-dimensional peripheral sensory input into structured population activity patterns. Each odour is characterised by a distinct chemical composition, and individual odorant molecules bind selectively to olfactory receptors expressed by olfactory sensory neurons (OSNs) located in the nose. This receptor-specific binding produces a combinatorial code of OSN activity, such that each odour evokes a characteristic spatial pattern of activation across the OB (Figure 4). Importantly, OSNs expressing the same receptor type converge onto a common glomerulus, a region within the OB where OSN axons terminate and the first stage of neural processing of smells occurs.

Within each glomerulus, OSN axon terminals form excitatory synapses onto multiple downstream neuronal populations. In addition to main signalling pathways, the glomerular layer also contains a number of inhibitory and excitatory interneurons that shape signal processing at this stage of sensory integration. These interneurons play a crucial role in gain control, contrast enhancement, and decorrelation of odour representations,

and are therefore central to both sensory computation and homeostatic regulation.

For modelling purposes, we group OB interneurons into functionally defined populations that capture their dominant circuit roles. This abstraction allows us to retain essential features of the circuit's connectivity without introducing an excessive number of parameters or variables. We acknowledge that real interneuron populations are diverse, and that this grouping averages over subtype-specific differences. Its validity rests on the assumption that dominant circuit-level dynamics are shaped by the main connectivity patterns rather than by small differences between individual cell types.

- **Anaxonic interneurons (blue, b):** These cells, which represent the periglomerular cell population,[11] drive local inhibition. By providing feedback inhibition within a single glomerulus, they regulate the level of excitation driven by OSN input. Functionally, they act as a rapid stabilising mechanism that prevents runaway excitation and contributes to local gain control. In the model, this population implements self-inhibitory feedback that helps stabilise glomerular activity.

- **Axonic interneurons (green, g):** These neurons, which represent short axon cells,[11] provide lateral inhibition across glomeruli. They receive excitatory input from one glomerulus and project inhibitory outputs to neighbouring glomeruli, enabling competitive interactions between odour representations. This lateral inhibitory network enhances contrast between similar odours by suppressing shared activity across neighbouring glomeruli, producing more distinct and separable responses at the circuit output.

- **External tufted cells (ETCs, red, r):** ETCs act as fast excitatory relays within the glomerular circuit, transmitting OSN-driven signals to mitral cells.[11] In the biological system, OSNs expressing the same olfactory receptor type converge onto two bilateral "sister" glomeruli, which are linked via ETC-mediated interactions. For modelling simplicity, these sister glomeruli are merged into a single representative unit, resulting in one ETC population per glomerulus.

- **Mitral cells (grey):** Mitral cells serve as the primary projection neurons of the OB, integrating input from glomerular circuits and transmitting processed olfactory information to higher cortical areas.[11] They receive both direct excitatory drive and indirect inhibitory modulation via interneuron networks, and therefore their activity reflects the net output of the olfactory bulb circuit.

- **Granule cells (black):** Granule cells provide dendrodendritic feedback inhibition onto mitral cells and are among the most numerous interneurons in the OB.[11] They integrate inputs from both mitral cells and cortical projections, suggesting they play an important role in experience-dependent modulation. Together with the anaxonic interneurons in the glomerular layer, granule cells are one of the few neuronal populations in the adult brain to undergo continuous neurogenesis, and their inhibitory feedback plays an important role in shaping mitral cell output.[14]

For convenience, we denote anaxonic interneurons as blue (b), axonic interneurons as green (g), and ETCs as red (r), following the colour scheme in Figure 4. This notation is used throughout the mathematical formulation of the model.

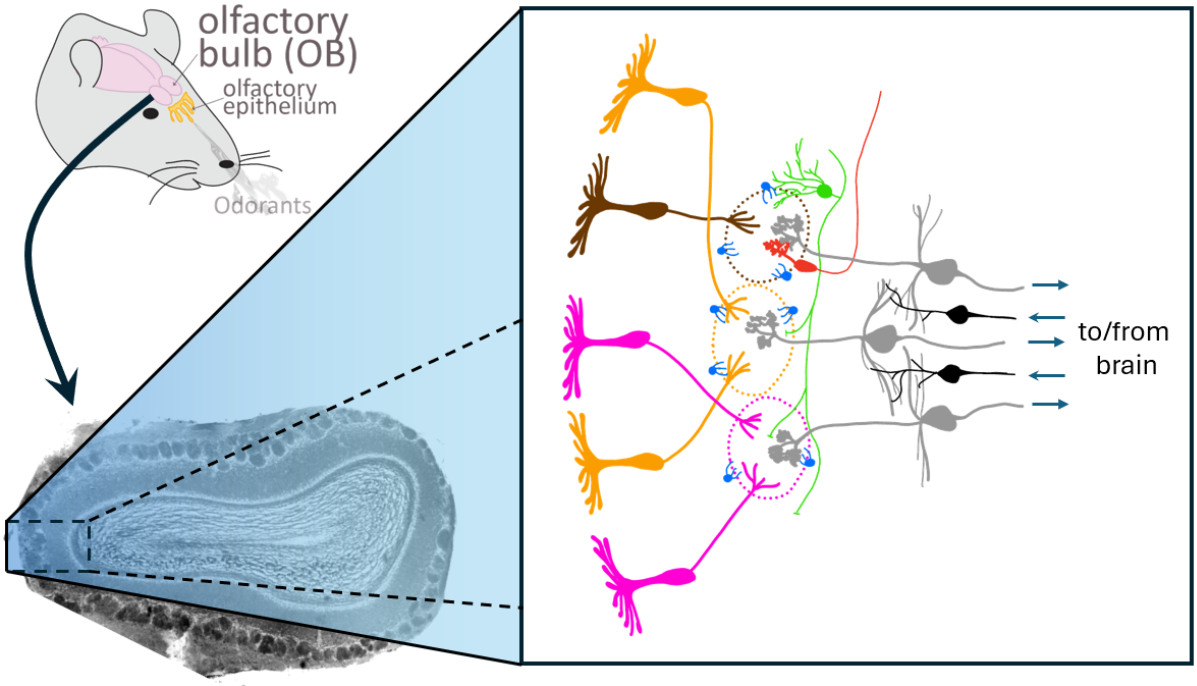


Figure 4: Simplified OB circuitry: OSNs (pink/orange/brown), axonic interneurons (green), anaxonic interneurons (blue), ETCs (red), mitral cells (grey), granule cells (black).

3 Mathematical Model

3.1 Model: Neuron

Figure 5 illustrates the way the interaction between two neurons is computed in the model. The first neuron (yellow) receives a stimulus x and generates an output z through a function σ_z, such that $z = \sigma_z(x)$. A fraction of this output, $w_z\ z$ (with $0 < w_z < 1$) serves as an input to the second neuron (purple), which applies its own function σ_y to generate the output $y = \sigma_y(w_z\ z)$.

In the model, all variables represent steady-state firing rates, which implicitly assumes slow homeostatic timescales relative to fast action potential dynamics. Synaptic weights w_i encode the effective strength of connections between neurons, while the functions σ_j capture the input-output response of each population. This allows us to focus on the input-output relationships at the circuit level, setting aside detailed biophysical mechanisms such as ion channel kinetics or subthreshold membrane dynamics.

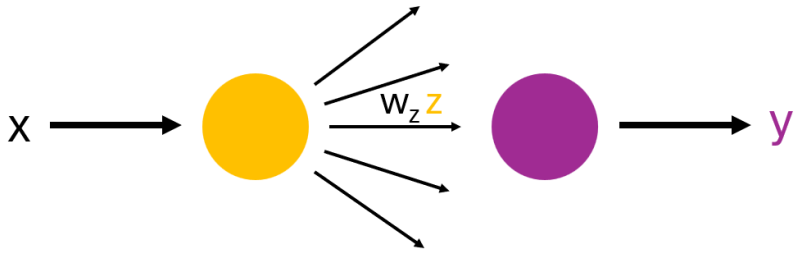


Figure 5: Two-neuron circuit model: x, z, y are steady-state firing rates and w denotes synaptic coupling strength. $z = \sigma_z(x)$ and $y = \sigma_y(w_z z)$

Neuronal response is modelled using the Naka–Rushton function (Equation (2)), which provides a simple, differentiable, and biologically motivated description of saturating nonlinear firing rate responses.[13] It is widely used in computational neuroscience as a firing-rate transfer function and captures the sigmoidal nonlinearity characteristic of real neurons. Firing rates are expressed in normalised units, with the maximum response set to 1:

$$\sigma_j(x) = \frac{x^{n_j}}{x^{n_j} + k_j^{n_j}} \tag{2}$$

Here, k_j is the semi-saturation constant, defining the input level at which the neuron reaches half its maximum response, while n_j (the Hill coefficient) determines how sharply the response transitions from low to high firing rates. Larger n_j produces a more switch-like response profile, whereas smaller values produce smoother, more gradual responses. To ensure physiologically plausible behaviour, all populations are kept within their dynamic range, avoiding extremes of silence or saturation.

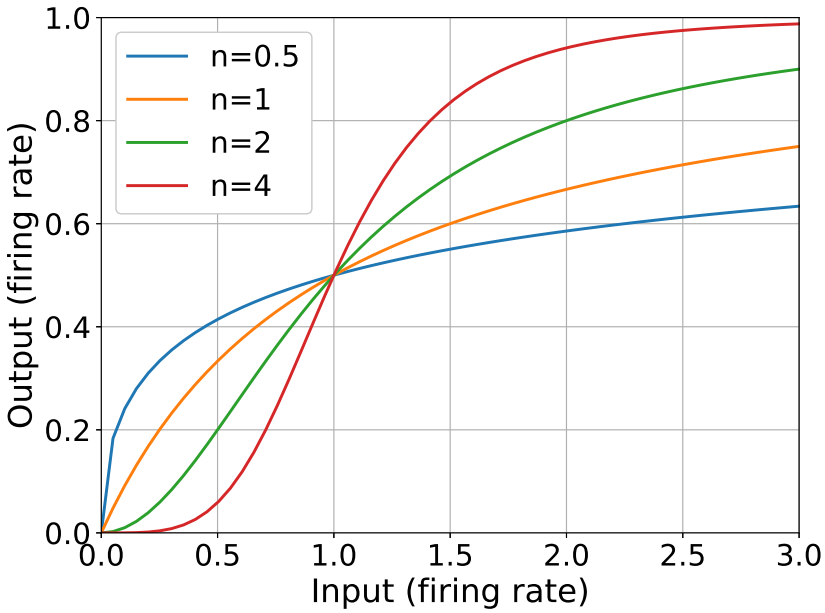


Figure 6: Naka–Rushton functions showing how the Hill coefficient n controls response steepness (Equation (2) with k=1).

3.2 Model: Setting the Hill Exponent n

In the circuit model, each neuronal population passes its input through a Naka–Rushton function (Equation (2)). As described in the previous section, the Hill exponent n governs the shape of this input–output relationship. Therefore, the choice of n directly determines how sensitively each cell type responds to changes in its input and how sharply it saturates. Since different neuronal populations in the olfactory bulb exhibit different response profiles, reflecting differences in ion channel composition and intrinsic excitability, it is biologically motivated and necessary to assign a distinct value of n to each cell type rather than using a single shared exponent.

3.2.1 Data Source

Each Hill exponent n_j was estimated by fitting the Naka-Rushton function to *in vitro* experimental firing-rate curves, in which the firing rate (Hz) was measured as a function of injected current (pA), which serves as a proxy for synaptic drive.[8] Recordings of unperturbed animals are used in this fitting, as the goal at this stage is to characterise the baseline input-output properties of each cell type under normal conditions. The experimental data used for fitting are shown in Figure 7.

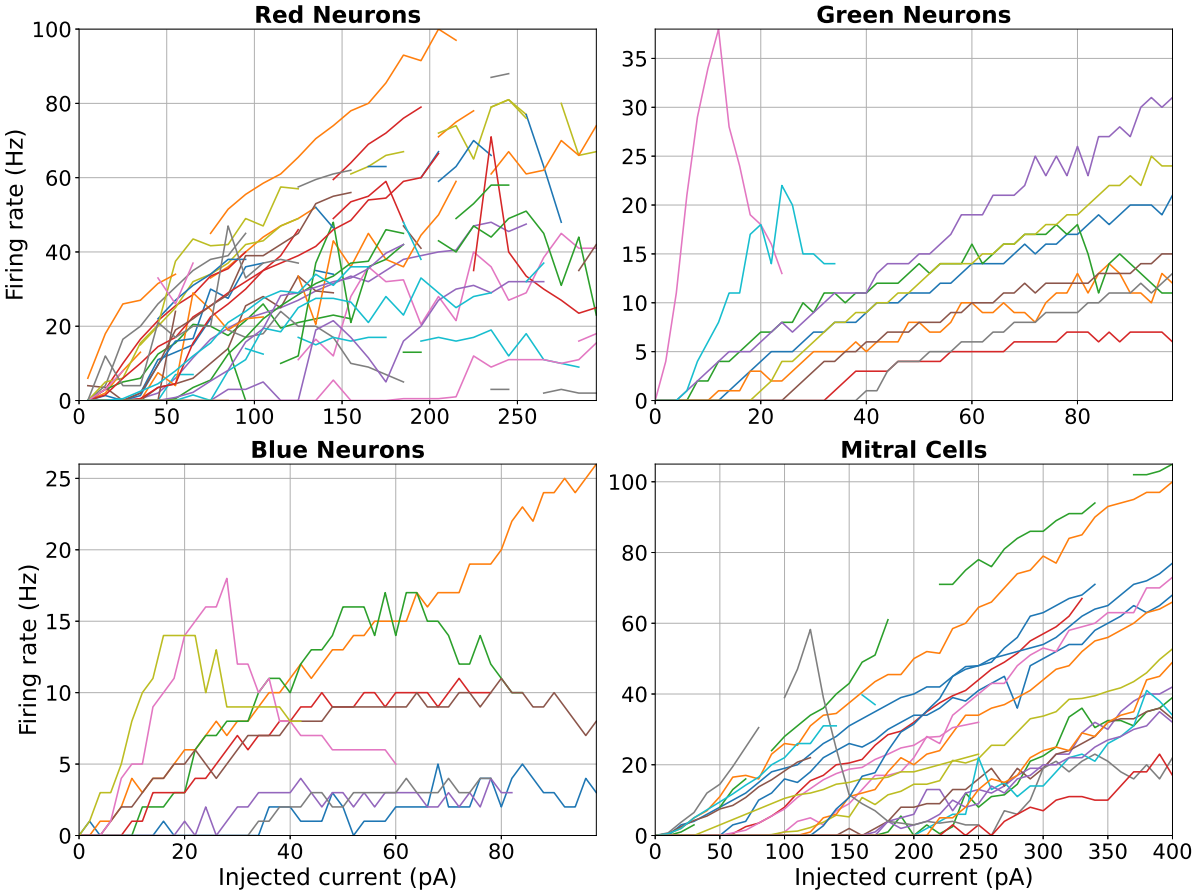


Figure 7: Firing-rate curves from unperturbed mice.[8] Each coloured line represents an individual cell of a given type. Firing rate (Hz) is recorded as a function of injected current (pA).

3.2.2 Fitting Procedure

For each cell class, a Naka–Rushton function (Equation (3)) was fitted to the mean firing rate $R(x)$ as a function of injected current x, where the firing rate represents population activity and the injected current represents the level of synaptic drive received by that population.

$$R(x) \;=\; y_{\max}\,\frac{x^{n}}{x^{n}+b^{n}} \tag{3}$$

Equation (3) has three free parameters: the Hill exponent n, which controls response steepness, the half-saturation constant $b > 0$, which sets the input level at which the cell reaches half its maximum firing rate, and the asymptotic firing rate $y_{\max}$, which scales the overall output. All three parameters were estimated jointly by nonlinear least squares optimisation.

A practical issue may arise when fitting non-integer exponents to the experimental data because real numbers raised to a non-integer power are undefined for negative values. For example, $(-1)^{0.5}$ has no real solution and evaluates to either `NaN` or a complex number, both without any physical meaning for our model. To prevent this, inputs are clipped to be non-negative before any power is taken:

$$x \;\leftarrow\; \max(x,\,0) \tag{4}$$

This is both numerically necessary as well as biologically sensible, since firing rates cannot be negative. Therefore, the clipping does not distort the fitted curve, it simply ensures that the optimisation remains well-defined throughout.

3.2.3 Fitted Exponents

The Naka–Rushton function provides a good description of the firing-rate curves across the cell populations, as shown in Figure 8. For all cell types, the fitted curves track the experimental average closely.

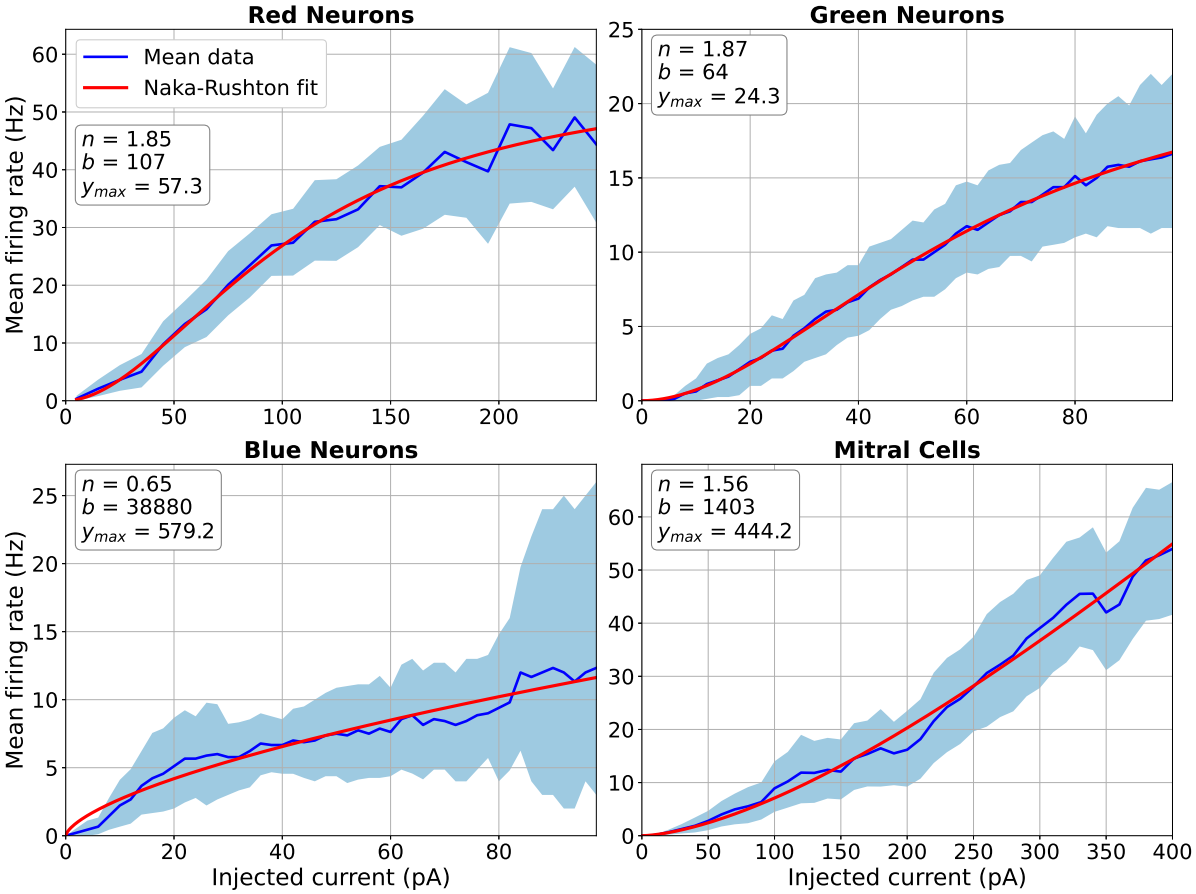


Figure 8: Naka–Rushton fitted to unperturbed firing-rate curves. Each panel shows the mean cell responses and 95% confidence intervals across all recorded cells of that type (average from Figure 7) together with the fitted curve (Equation (3)) with the inferred Hill exponent n, half-saturation constant b, and maximum firing rate y_{max}.

Three out of the four Hill exponents are larger than 1, which result in curves with a sigmoidal shape. The exception is the blue (anaxonic) interneuron population, for which $n \approx 0.65 < 1$, producing a monotonically increasing response with no inflection point.

3.2.4 Reparameterisation of Fitted Functions

For computational efficiency, all inputs are rescaled as $x' = x/b_j$, which normalises the semi-saturation point to unity across all populations:

$$\sigma_j(x') = \frac{{x'}^{n_j}}{{x'}^{n_j} + 1} \tag{5}$$

This rescaling is a valid reparameterisation, as the semi-saturation constant b is not removed from the model but rather absorbed into the input weight w, which is now permitted to take any positive value. This reparameterisation does not alter what the model can represent, it simply reduces the number of free parameters it contains and simplifies its numerical implementation. Figure 9 compares the reparameterised Naka–Rushton functions (Equation (5)) for the four cell types using the experimentally fitted Hill exponents n_j.

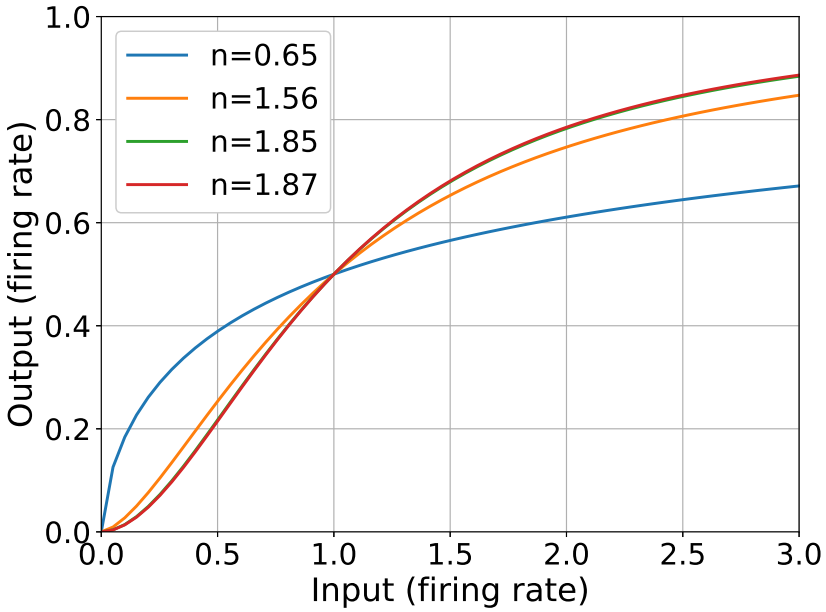


Figure 9: Naka–Rushton function with fixed n for each population (normalised Equation (5)). Steeper curves correspond to larger n.

3.3 Model: Olfactory Bulb

With the key neuron types and their connectivity established, we now construct a mathematical model of the OB using the neuronal input–output model described in Section 2.2.

The model is built on several simplifying assumptions, each motivated by the goal of capturing essential circuit behaviour while keeping the number of free parameters manageable:

3.3.1 Model Assumptions

- **One representative cell per population:** Rather than simulating every individual neuron, each cell class is represented by a single unit whose activity reflects the average firing rate of that population. This approximation is standard in circuit-level modelling and is appropriate here because we are interested in the collective behaviour of each cell type, i.e. how strongly a population is driven and how it influences others, rather than cell-to-cell interactions.

- **Two glomeruli:** A single glomerulus would suffice if all inhibitory interactions were local, but the axonic (green, see Figure 4) interneurons project laterally across glomeruli, providing cross-glomerular inhibition that is an important feature of OB circuit. Having two glomeruli is the minimal architecture that captures this inter-glomerular coupling while keeping the model compact. Anaxonic (blue) interneurons, by contrast, are assumed to inhibit only within their own glomerulus.

- **Steady-state firing rates:** Since the aim is to study homeostatic plasticity, the model focuses on steady-state firing rates rather than transient dynamics. Temporal differential equations are therefore not required. Instead, the model defines a set of algebraic constraints that the firing rates must satisfy at equilibrium. This substantially reduces the complexity of the model without sacrificing the features relevant to homeostatic regulation.

- **Symmetric weight sharing:** To further reduce the number of free parameters, corresponding synaptic connections are constrained to be identical across the two glomeruli. For example, the weight from blue to green neurons in glomerulus 2 is set equal to that in glomerulus 1. This symmetry assumption is reasonable given that the two glomeruli are treated as structurally equivalent, and it ensures that differences in activity between glomeruli arise only from differences in sensory input rather than from asymmetric connectivity.

3.3.2 Circuit Equations

Under the assumptions above, the biological connectivity sketched in Figure 4 is reduced to the circuit shown in Figure 10, where each arrow in the schematic represents a synaptic connections. The algebraic form of these connections is given below.

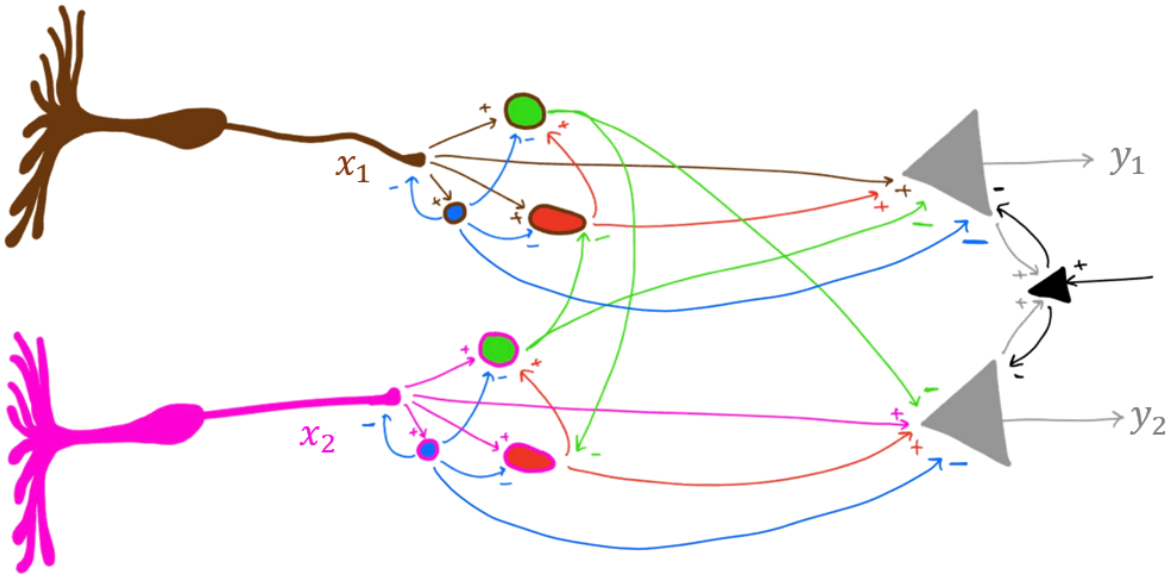


Figure 10: Simplified OB circuit used in the model. Excitatory (+) and inhibitory (-) connections between the blue (b), red (r), green (g), mitral (y), and granule (gr) populations follow the architecture of Figure 4 after applying the assumptions.

The steady-state firing rates of the blue (b), red (r), green (g), mitral (y), and granule (gr) cell populations are given by:

$$b = \sigma_b \left(w_{\text{osn},b}\, \alpha x \right) \tag{6}$$

$$r = \sigma_r \left(w_{\text{osn},r}\, \alpha x - w_{b,r}\, b - \mathbf{W}_{g,r}\, g \right) \tag{7}$$

$$g = \sigma_g \left(w_{\text{osn},g}\, \alpha x - w_{b,g}\, b + w_{r,g}\, r \right) \tag{8}$$

$$y = \sigma_y \left(w_{\text{osn},y}\, \alpha x - w_{b,y}\, b - \mathbf{W}_{g,y}\, g + w_{r,y}\, r - w_{gr,y}\, gr \right) \tag{9}$$

$$gr = \sigma_g r \left(w_{\text{cor},gr}\, \text{cortex} + w_{y,gr}\, y \right) \tag{10}$$

Here σ_j denotes the Naka–Rushton function (Equation (5)), which initially are assumed to have fixed population-specific n values. Each weight $w_{i,j}$ encodes the synaptic strength of the connection from population i to population j, and is rescaled by the semi-saturation constant of the postsynaptic population j to account for the reparameterisation described in Section 3.2.

The parameter $\alpha \in [0, 1]$ accounts for feedback from anaxonic (blue) interneurons onto the OSN. Therefore, in the steady-state equations, the downstream populations only see a fraction α of the initial sensory input x. In the initial configuration, the sensory input to both glomeruli is set to $x = 1$.

Connections that are local to a glomerulus appear as scalar weights (e.g. $w_{b,g}$). However, connections from green interneurons to red or mitral cells in the other glomerulus are represented by 2×2 matrices $\mathbf{W}_{g,r}$ and $\mathbf{W}_{g,y}$:

$$\mathbf{W}_{g,r} = \begin{bmatrix} 0 & w_{g_1,r_2} \\ w_{g_2,r_1} & 0 \end{bmatrix} \qquad \mathbf{W}_{g,y} = \begin{bmatrix} 0 & w_{g_1,y_2} \\ w_{g_2,y_1} & 0 \end{bmatrix} \tag{11}$$

where the zero diagonal entries reflect the absence of intra-glomerular projections from green cells, and $w_{g_1,r_2} = w_{g_2,r_1}$ and $w_{g_1,y_2} = w_{g_2,y_1}$ reduce each matrix to a single free parameter.

3.4 Model: Fixed-Point Iteration

The circuit equations (Equations (6)–(10)) form a nonlinear system in which the firing rates of all populations are coupled through recurrent connections, for which there is no obvious analytical solution. We therefore solve them numerically as fixed-point problems, identifying activity states in which each population's output is self-consistent with the inputs it receives from the rest of the circuit.

Simple fixed-point iteration can converge slowly or fail when recurrent coupling is strong. Therefore, we use Anderson acceleration,[15] which improves convergence by forming the next iterate as a linear combination of several recent updates rather than relying solely on the most recent one. Intuitively, this allows the solver to extrapolate from the trend of previous iterates, damping oscillations that would otherwise slow or prevent convergence. Typically, we use a history of 5 iterates, a convergence tolerance of 10^{-10}, and a maximum of 50 iterations, which reliably finds the fixed point.

3.4.1 Intra-glomerular loop (r and g)

Equations (7) and (8) form a coupled fixed-point problem in which the red (ETC) population provides excitatory drive to the green (axonic) population, which in turn projects inhibitory feedback onto the ETCs. This reciprocal coupling is resolved using

Anderson acceleration, with the hyperparameters described above.

3.4.2 Mitral–granule loop (y and gr)

Equations (9) and (10) form a second fixed-point problem. Mitral cell activity drives granule cells through excitatory synapses, while granule cells in turn inhibit mitral cells via dendrodendritic feedback, creating a second recurrent loop. In principle, the same Anderson acceleration scheme could be applied here.

However, the granule cell equation (Equation (10)) includes a cortical input term that is both highly variable and poorly constrained by available experimental data. Current evidence suggests that cortical projections to granule cells are more closely associated with context-dependent modulation and associative learning than with the homeostatic regulation, the focus of this project.[3]

Including an unconstrained cortical drive would therefore introduce a free parameter that could not be reliably estimated and might obscure the homeostatic mechanisms of interest. For these reasons, granule cells are excluded from the circuit analysis.

3.5 Model: Parameter Initialisation

Before modelling the effects of sensory deprivation, we must first identify a set of parameters $\theta = \{w_{osn,r}, w_{osn,b}, w_{osn,g}, w_{osn,y}, w_{b,r}, w_{b,g}, w_{b,y}, w_{r,y}, w_{r,g}, \alpha, W_{g,r}, W_{g,y}\}$ that produces a plausible baseline state of the circuit, one that is consistent with normal, unperturbed OB function. Two conditions must be satisfied simultaneously, the mitral cell output y should match a prescribed target value, and the firing rates of all populations should be within their dynamic range.

Regarding the second condition, the Naka–Rushton function saturates at both ends. Near zero input, changes in drive produce almost no change in firing rate, and similarly near saturation. A neuron operating at either extreme is effectively unable to respond to changes in its input and therefore conveys little information about the circuit state.

From a homeostatic perspective, this is problematic as a saturated population has no capacity to increase its output further, and a silent population cannot reduce its output. Neurons operating near the midpoint of their dynamic range, around a normalised firing rate of 0.5, are maximally sensitive to input changes and retain the capacity to both increase and decrease their output in response to perturbations. Therefore, for a nominal input $x = 1$, we set 0.5 as the target firing rate for all cell populations.

Finding parameters that satisfy these conditions is formulated as a least-squares optimisation problem, where the loss function penalises deviations of each population's firing rate from the target value of 0.5:

$$L(\theta) = \sum_{i=1}^{m} \Big[(r_i - r_{t,i})^2 + (g_i - g_{t,i})^2 + (b_i - b_{t,i})^2 + (y_i - y_{t,i})^2 \Big] \tag{12}$$

where $m = 2$ is the number of glomeruli, and r_i, g_i, b_i, y_i are the steady-state firing rates of the red, green, blue, and mitral cell populations, respectively, and $r_{t,i} = g_{t,i} = b_{t,i} = y_{t,i} = 0.5$ are the target rates for all i. To ensure biological plausibility, synaptic weights are clipped to be non-negative and α is restricted to $[0, 1]$ throughout the optimisation.

We minimise $L(\theta)$ using the Adam optimiser, which adapts the learning rate for each parameter individually using exponentially weighted averages of past gradients ($\beta_1 = 0.9$) and squared gradients ($\beta_2 = 0.999$), with a learning rate of $\alpha = 0.001$.[16] This adaptive scheme is more robust than standard gradient descent when parameters differ in scale or the loss landscape is highly nonlinear, both of which are likely given the circuit structure. The algorithm is implemented in JAX/Optax with JIT compilation, which enables efficient automatic differentiation through the fixed-point solver.[17].

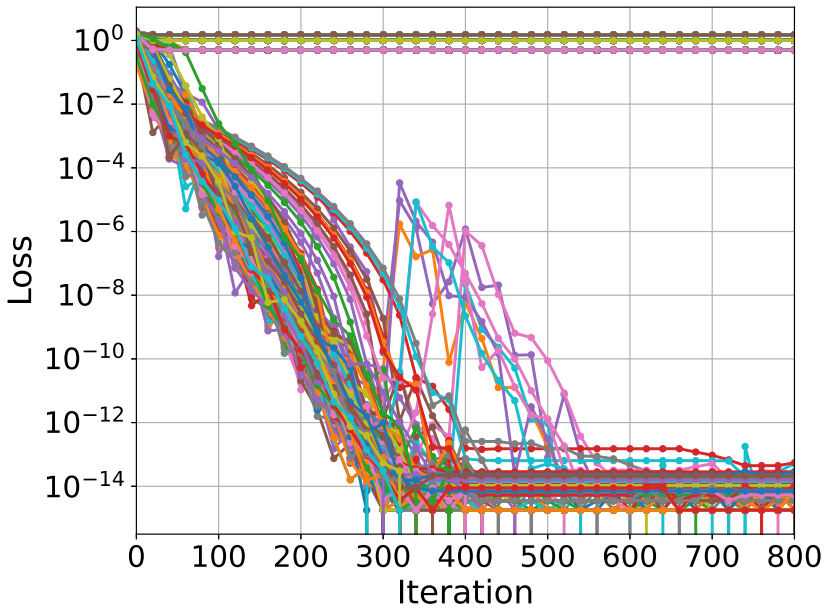


Figure 11: Loss $L(\theta)$ during Adam optimisation across 1000 random initialisations, with each coloured line representing a different initialisation. Curves that reach near-zero ($< 10^{-12}$) loss correspond to the 570 parameter sets retained for later analysis. The remaining 430 initialisations failed to converge and are discarded.

Optimised parameter sets are retained only if the outputs of all cell types fall within 0.5 ± 0.001. Those outside this range are discarded. Of the 1000 random initialisations, 570 satisfy this criterion and are carried forward for subsequent analysis.

As shown in Figure 11, the loss converges reliably across the 570 retained initialisations. However, the optimised parameters are not unique. Figure 12 shows the distribution of raw parameter values across all retained solutions. Many different combinations of synaptic weights and α values can produce the required target firing rate. This degeneracy is a consequence of the circuit having more free parameters than constraints, a situation that is unavoidable in a model of this complexity. Rather than treating this as a problem, we embrace it. The full set of valid parameter solutions forms a family of circuits that all satisfy the same criteria. We can use this set to ask which properties are shared across solutions, and are therefore robust features of the model.

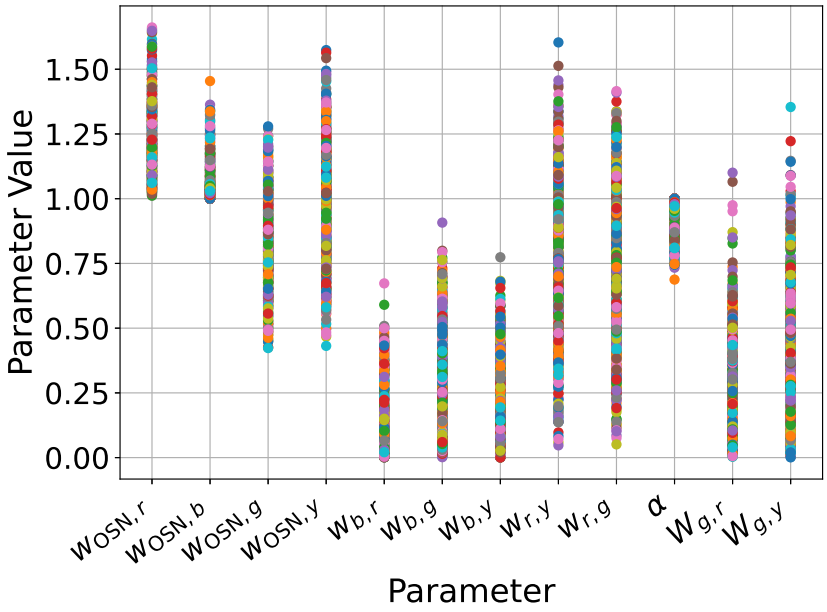


Figure 12: Raw parameter values across all 570 retained solutions. The spread across solutions reflects the degeneracy of the optimisation problem. Each colour represents a different retained parameter set.

3.6 Model: Stability Analysis

The optimisation procedure in the previous section identifies parameter configurations that satisfy the target firing-rate conditions at steady state. However, satisfying the fixed-point equations is a necessary but not sufficient condition for biological plausibility. In a real neural circuit, activity is subject to continuous perturbation from noise, fluctuating inputs, and ongoing network dynamics. A biologically viable operating point must therefore not only exist as a fixed point but also be stable, i.e. small deviations from the steady state should decay over time rather than grow. We therefore assess the local stability of each identified fixed point before using it in further analyses.

3.6.1 Network dynamics model

To assess stability, we require a model of how firing rates evolve over time. We adopt the standard continuous-time firing-rate equation:[13]

$$\tau \frac{ds}{dt} = -s + F(s, \theta, x) \tag{13}$$

where $s(t) = [r(t), g(t), b(t), y(t)]^\top$ is the vector of population firing rates, θ are the synaptic weight parameters, x is the external sensory input, and $\tau > 0$ is the characteristic time constant governing how quickly firing rates respond to changes in input. The term $-s$ represents the passive decay of activity in the absence of drive, ensuring that firing rates return to zero if all inputs are removed.

The function F describes the network response obtained from the circuit equations (Equations (6)–(9)), evaluated using the current firing rates $s(t)$ rather than their steady state values:

$$F(s,\theta,x) = \begin{bmatrix} F_r(s,\theta,x) \\ F_g(s,\theta,x) \\ F_b(s,\theta,x) \\ F_y(s,\theta,x) \end{bmatrix} = \begin{bmatrix} \sigma\left(w_{\text{osn},r}\,\alpha x - w_{b,r}\,b - \mathbf{W}_{g,r}\,g\right) \\ \sigma\left(w_{\text{osn},g}\,\alpha x - w_{b,g}\,b + w_{r,g}\,r\right) \\ \sigma\left(w_{\text{osn},b}\,\alpha x\right) \\ \sigma\left(w_{\text{osn},y}\,\alpha x - w_{b,y}\,b - \mathbf{W}_{g,y}\,g + w_{r,y}\,r\right) \end{bmatrix} \tag{14}$$

A steady-state solution s^* satisfies $ds/dt = 0$, and therefore, Equation (13) implies that fixed points must satisfy:

$$F(s^*,\theta,x) = s^* \tag{15}$$

3.6.2 Linearisation

To determine whether s^* is stable, we analyse the behaviour of small perturbations around it. Writing $s(t) = s^* + \delta(t)$, where $\delta(t)$ represents a small deviation from the fixed point, and substituting into Equation (13):

$$\tau\frac{d\delta}{dt} = -(s^* + \delta) + F(s^* + \delta,\, \theta,\, x) \tag{16}$$

Expanding F in a first-order Taylor series around s^*:

$$F(s^* + \delta,\, \theta,\, x) \approx F(s^*,\, \theta,\, x) + \mathbf{J}\,\delta \tag{17}$$

where $\mathbf{J} = \partial F/\partial s\big|_{s^*}$ is the Jacobian of F evaluated at the fixed point. Substituting Equation (17) in Equation (16), applying the fixed-point condition $F(s^*,\theta,x) = s^*$ and simplifying we obtain the linear differential equation governing the evolution of perturbations around the fixed point:

$$\frac{d\delta}{dt} = \frac{1}{\tau}(\mathbf{J} - \mathbf{I})\,\delta = \mathbf{A}\,\delta \tag{18}$$

where $\mathbf{A} = (\mathbf{J} - \mathbf{I})/\tau$ is the linearised system matrix. This is a linear ordinary differential equation whose solution is $\delta(t) = e^{\mathbf{A}t}\delta(0)$. Therefore, the fixed point is locally asymptotically stable if and only if all eigenvalues of $\mathbf{A}$ have strictly negative real parts, so that perturbations decay exponentially to zero. Since $\tau > 0$ rescales all eigenvalues uniformly without changing their signs, stability depends entirely on the eigenvalues of $\mathbf{J} - \mathbf{I}$.

3.6.3 Stability Results

Figure 13 shows the real parts of the eigenvalues of $\mathbf{A}$ across all 570 retained parameter configurations. All eigenvalues have strictly negative real parts, confirming that every identified fixed point is locally stable. This provides confidence that the baseline parameter solutions represent viable operating points that the circuit could maintain in the presence of small perturbations.

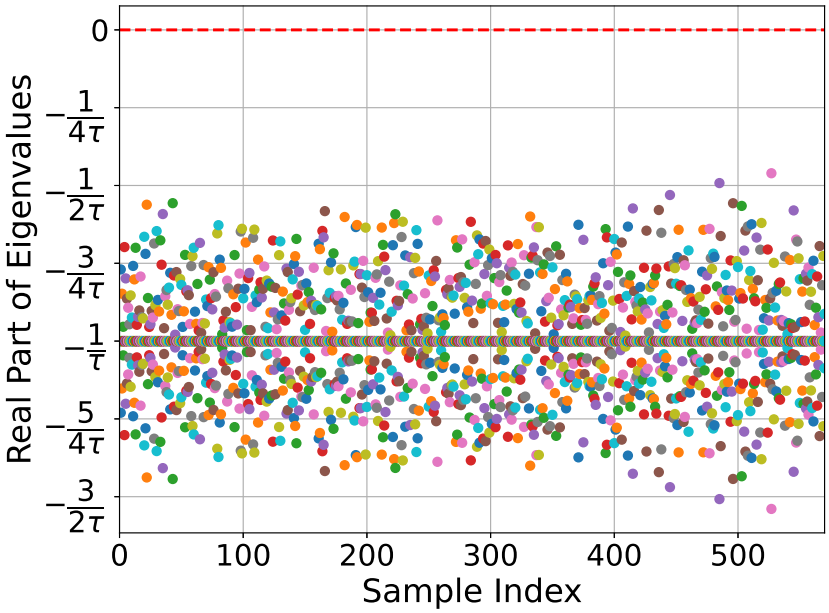


Figure 13: Real parts of the eigenvalues of **A** across all 570 retained parameter configurations. All values are strictly negative, confirming local asymptotic stability at every identified fixed point.

Many eigenvalues lie close to $-1/\tau$, which follows from the structure of the linearised system. Each neuronal population depends only on a subset of the network variables (Equations (7)–(9)), so Jacobian **J** is sparse and its off-diagonal coupling terms are generally small. Consequently, the dominant contribution to the dynamics comes from the decay term $-s$ in Equation (13), which contributes the diagonal term $-\mathbf{I}/\tau$ in Equation (18). Along weakly coupled directions of state space, the eigenvalues therefore remain close to $-1/\tau$.

4 Homeostatic Plasticity (HP): $\Delta\theta$ Preserving Output

The sensory deprivation experiment described in Section 2.1 raises the following question: how does the olfactory bulb circuit maintain stable output despite the axon initial segment (AIS) shrinkage in the green neurons induced by sensory deprivation? Given that dopaminergic axonic interneurons exhibit a reduction in AIS length and therefore provide less drive to mitral cells, the preservation of stable mitral cell output suggests that compensatory mechanisms must emerge elsewhere in the circuit.

Formally, we seek to characterise the set of parameter changes $\Delta\theta$ that leave the circuit output y unchanged. In other words, we want to identify directions in parameter space along which the system can evolve without altering mitral cell output.

4.1 HP: Taylor Expansion of the Output

To characterise output-preserving parameter changes analytically, we expand the mitral cell output y as a function of parameters in a second-order Taylor series around the previously optimised baseline parameter sets θ_o from Section 3.5:

$$y(\theta_o + \Delta\theta) \approx \; y(\theta_o) + \text{grad}^T \Delta\theta + \frac{1}{2}\Delta\theta^T \mathbf{H}\, \Delta\theta + \mathcal{O}(\|\Delta\theta\|^3) \tag{19}$$

where y is the mitral cell readout, $\text{grad} = \nabla_\theta y(\theta_o)$ is the output gradient with respect to the parameters, and $\mathbf{H} = \nabla^2_\theta y(\theta_o)$ is the corresponding Hessian matrix. Both grad and $\mathbf{H}$ are evaluated at each baseline initialisation θ_o and computed via automatic differentiation through the fixed-point solver using JAX.

The condition for the output to remain unchanged after a (small) parameter perturbation, $y(\theta_o + \Delta\theta) \approx y(\theta_o)$, reduces to:

$$\text{grad}^T \Delta\theta + \frac{1}{2}\, \Delta\theta^T \mathbf{H}\, \Delta\theta \approx 0 \tag{20}$$

Equation (20) can be satisfied in two ways. In the first scenario, the first- and second-order contributions cancel each other out, so that perturbations with a nonzero linear effect on the output are offset by curvature-induced corrections, producing finely tuned parameter combinations that preserve the output. Biologically, this corresponds to a regime of delicate compensation requiring coordinated adjustments across parameters and precise alignment between perturbation directions and local curvature in parameter space. These tight requirements make this regime an unlikely approach for robust biological regulation.

In the second scenario, both the first- and second-order contributions vanish to zero. The linear term vanishes whenever the perturbation $\Delta\theta$ is orthogonal to the gradient, i.e. when $\text{grad}^T \Delta\theta = 0$. The quadratic term vanishes when $\Delta\theta$ aligns with eigenvectors of the Hessian associated with near-zero eigenvalues, i.e. locally flat curvature directions. Together, these conditions define a locally output-invariant subspace in parameter space within which synaptic weights can vary without significantly affecting circuit output. Assuming such a subspace exists, this regime offers a more robust route to output invariance and is therefore a more plausible candidate for homeostatic regulation.

We therefore focus on this second scenario and identify the output-invariant subspace

as the set of parameter directions $\Delta\theta$ that satisfy two conditions: they are orthogonal to the gradient ($\text{grad}^T \Delta\theta = 0$), and they align with eigenvectors of the Hessian whose eigenvalues are approximately zero ($|\lambda_i| \approx 0$). Moving along any direction in this subspace leaves the circuit output unchanged.

4.2 HP: Computing the Hessian

Having established that output-preserving parameter changes lie along directions that are both perpendicular to the gradient grad and aligned with eigenvectors of the Hessian $\mathbf{H}$ with near-zero eigenvalues (Section 4.1), we now compute $\mathbf{H}$ explicitly for each of the baseline initialisation parameter set identified in Section 3.5.

Because the circuit contains two glomeruli, the mitral cell output is a two-dimensional vector (y_1, y_2). To define a scalar-valued Hessian, we let the readout to be $y = y_1 + y_2$ and compute:

$$\mathbf{H} = \nabla^2_\theta y(\theta_o) \tag{21}$$

This choice is natural given the symmetric structure of the two-glomerulus model, i.e. both mitral cells are treated equivalently, and their summed output captures the total network drive to downstream cortical areas. The Hessian is computed via automatic differentiation using reverse-mode Jacobians in JAX ($H =$ `jax.jacrev(jax.grad(f))`), which is efficient and numerically stable for scalar objectives. This avoids the fragility of pushing forward-mode through fixed-point solvers, where implicit differentiation is used.

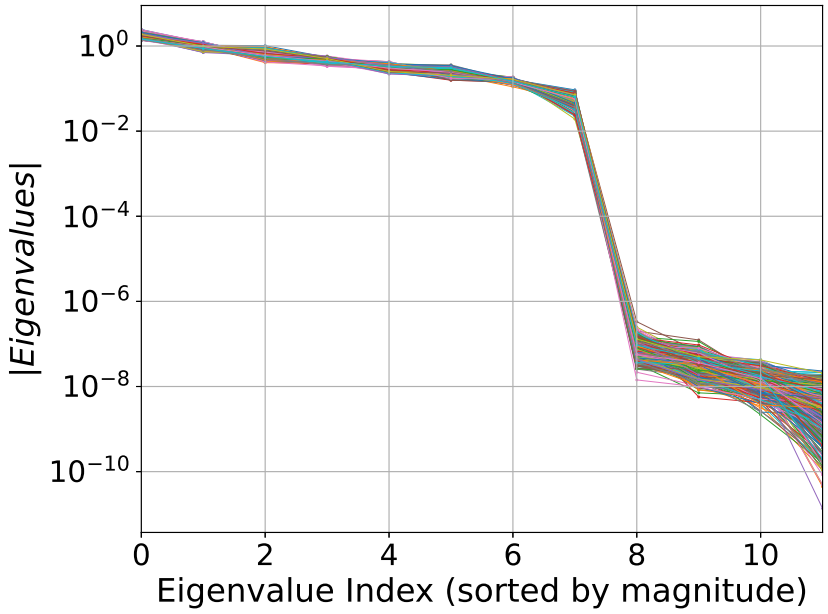


Figure 14: Distribution of eigenvalue magnitudes $|\lambda_i|$ of the Hessian $\mathbf{H}$ across all retained baseline parameters. Four eigenvalues are consistently small relative to the rest, revealing a near-null subspace of dimension four.

Figure 14 shows the distribution of Hessian eigenvalue magnitudes $|\lambda_i|$ across all retained parameter sets. A consistent structure emerges. Four eigenvalues are substantially smaller than the other eight, indicating the presence of a near-null subspace of dimension four in parameter space. Perturbations along the corresponding eigenvectors change the output very little (to second order in $\Delta\theta$) making them natural candidate directions for homeostatic compensation.

The next step is to ask whether these directions are consistent across parameter sets. In other words, whether different parameter configurations agree on which weight changes leave the output unchanged. If this is the case, this would suggest that the output-preserving subspace is a robust feature of the circuit structure.

4.3 HP: Common Subspace

For each retained baseline parameter, the Hessian analysis produces four eigenvectors that span a near-null subspace. Concatenating these across all $N = 570$ parameter sets produces a collection of ($2280 = 4 \times 570$) vectors in the twelve-dimensional parameter space. Directly analysing such a large collection is difficult, and there is no guarantee that the vectors from different solutions are aligned. Therefore, we seek a simpler representation of the common structure shared across all solutions.

To identify the dominant directions that appear most consistently across parameter sets, we apply singular value decomposition (SVD) to the $[12 \times 4N]$ matrix obtained by stacking all eigenvectors column-wise:

$$\mathbf{V} = \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{W}^T \tag{22}$$

The columns of $\mathbf{U}$ are the left singular vectors, which define orthonormal directions in parameter space, and the diagonal entries of $\mathbf{\Sigma}$ are the corresponding singular values. A large singular value indicates that a particular direction in parameter space recurs frequently across the collection of eigenvectors. This means that it is a direction many different independent parameter sets agree is output-preserving. On the other hand, small singular values correspond to directions that appear only sporadically.

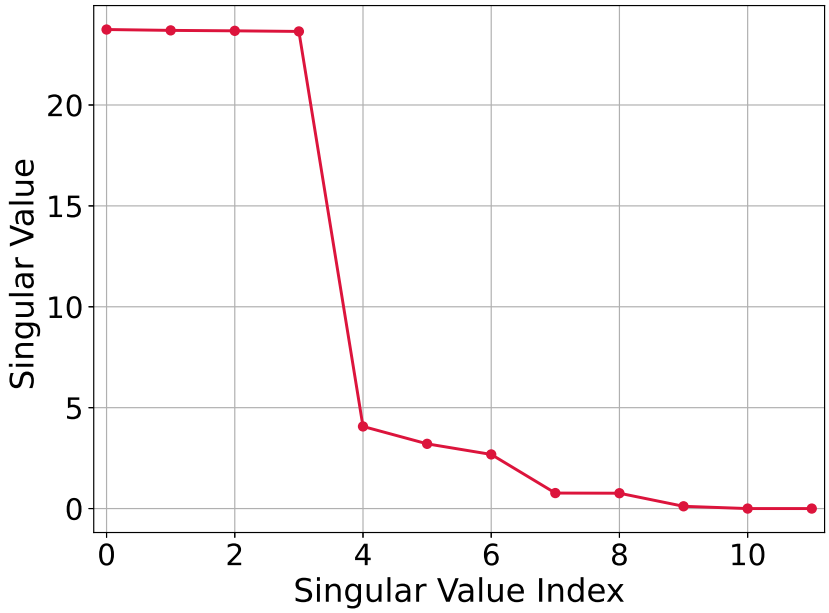


Figure 15: Singular values of the stacked near-null eigenvector matrix across all parameter sets.

Figure 15 shows the singular values. A clear structure is visible, where the first four singular values are substantially larger than the rest, suggesting that the near-null subspaces of different parameter solutions share a common four-dimensional structure. We take the four leading left singular vectors of $\mathbf{U}$ as a basis for this common output-preserving subspace.

This result is biologically significant because it suggests that the directions in synaptic weight space that preserve circuit output are not unique to any single parameter configuration. Instead, they appear to be a general property of the olfactory bulb network architecture.

The resulting basis vectors, however, are not yet in an interpretable form. Each is a dense linear combination of all twelve parameters, making it difficult to identify which synaptic connections are primarily involved. To address this, we apply a rotation to find the sparsest possible representation of the same subspace.

4.4 HP: Sparsification

The four basis vectors obtained from the SVD analysis from Section 4.3 span the output-preserving subspace, but as dense vectors in parameter space they offer limited biological interpretability. Each direction involves simultaneous changes across many synaptic weights. However, since the subspace is invariant under orthogonal rotation, we are able to choose a basis that is as sparse as possible while spanning the same space. A sparse basis is desirable because it would identify a small number of synaptic parameters that can vary independently to preserve output, providing clear and testable predictions about which connections are involved in homeostatic plasticity.

4.4.1 Optimisation Problem

Let $\mathbf{U} \in \mathbb{R}^{12\times 4}$ denote the matrix whose columns are the four leading left singular vectors. We seek a rotation that produces the sparsest possible basis spanning the same subspace. We introduce an orthogonal rotation matrix $\mathbf{Q} \in \mathbb{R}^{4\times 4}$ satisfying $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$ and define the rotated basis:

$$\hat{\mathbf{U}} = \mathbf{U}\mathbf{Q} \tag{23}$$

Since $\mathbf{Q}$ is orthogonal, the columns of $\hat{\mathbf{U}}$ span the same subspace as those of $\mathbf{U}$, so no information is lost. The optimisation problem is to find the rotation $\mathbf{Q}$ that produces the sparsest rotated basis, which we define by minimising the L_1 norm of $\hat{\mathbf{U}}$:

$$\min_{\mathbf{Q}} \mathcal{L}(\mathbf{Q}) = \|\hat{\mathbf{U}}\|_1 = \sum_{i=1}^{12}\sum_{j=1}^{4} \hat{\mathbf{U}}_{ij} = \sum_{i=1}^{12}\sum_{j=1}^{4} (\mathbf{U}\mathbf{Q})_{ij} \quad \text{subject to} \quad \mathbf{Q}^T\mathbf{Q} = \mathbf{I} \tag{24}$$

The L_1 norm promotes sparsity by penalising all nonzero entries equally, encouraging entries of $\hat{\mathbf{U}}$ to collapse toward zero. This produces new basis vectors that require only a small number of synaptic parameters, enhancing biological interpretability.

To enforce orthogonality efficiently, we first relax the constraint by introducing an unconstrained matrix $\tilde{\mathbf{Q}} \in \mathbb{R}^{4\times 4}$, which is free to take any value. To recover a valid orthogonal matrix from $\tilde{\mathbf{Q}}$, we apply a QR decomposition:

$$\tilde{\mathbf{Q}} = \mathbf{Q}\mathbf{R} \tag{25}$$

where $\mathbf{Q}$ is the orthonormal factor satisfying $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, and $\mathbf{R}$ is upper triangular. At each step we discard $\mathbf{R}$ and use only $\mathbf{Q}$, which re-imposes the orthogonality constraint. This converts the constrained optimisation over orthogonal matrices into an unconstrained optimisation over $\tilde{\mathbf{Q}}$, which can be solved with standard gradient-based methods.

4.4.2 Gradient Computation

Gradients are propagated through the computational flow:

$$\tilde{\mathbf{Q}} \xrightarrow{\text{QR}} \mathbf{Q} \xrightarrow{\hat{\mathbf{U}}=\mathbf{UQ}} \hat{u} \xrightarrow{\|\cdot\|_1} \mathcal{L} \tag{26}$$

Since the derivative of the absolute value function is

$$\frac{d}{dx}|x| = \text{sign}(x) = \begin{cases} +1, & x > 0 \\ -1, & x < 0 \end{cases} \tag{27}$$

the gradient of the loss with respect to $\hat{\mathbf{U}}$ is simply:

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{U}}_{ij}} = \text{sign}(\hat{\mathbf{U}}_{ij}) \tag{28}$$

This magnitude-independent gradient applies the same force to every nonzero entry of $\hat{\mathbf{U}}$ regardless of its magnitude, steadily driving all values toward zero and therefore promoting sparsity.

If we were to have used the L_2 norm, $\mathcal{L} = \sum_{ij} \hat{\mathbf{U}}_{ij}^2$, the gradient would be:

$$\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{U}}_{ij}} = 2\hat{\mathbf{U}}_{ij} \tag{29}$$

This gradient scales with magnitude and as a result, large entries are strongly shrunk, while small entries receive vanishing updates and typically remain nonzero. Therefore, L_2 norm encourages small-but-dense solutions, whereas L_1 promotes sparsity.

4.4.3 Optimisation using Adam

We minimise $\mathcal{L}(\mathbf{Q}(\tilde{\mathbf{Q}}))$ (Equation (24)) using the Adam optimiser applied to the unconstrained matrix $\tilde{\mathbf{Q}}$. At each iteration, we obtain an orthogonal $\mathbf{Q}$ from the QR decomposition $\tilde{\mathbf{Q}} = \mathbf{QR}$ and evaluate the loss on $\mathbf{Q}$. We use the Adam optimiser because the L_1 loss is non-smooth (its gradient is $\text{sign}(x)$, which is discontinuous at zero), and because the QR map from $\tilde{\mathbf{Q}}$ to $\mathbf{Q}$ introduces a nonlinear projection step. Adam's per-parameter adaptive learning rates help stabilise optimisation in both cases, whereas a fixed learning rate would struggle.

After optimisation, the final orthogonal matrix is recovered by applying QR decomposition to $\tilde{\mathbf{Q}}_{\text{final}}$:

$$\tilde{\mathbf{Q}}_{\text{final}} = \mathbf{Q}_{\text{opt}}\mathbf{R} \tag{30}$$

Solving for the orthogonal component $\mathbf{Q}_{\text{opt}}$, we define the sparse basis as:

$$\mathbf{U}_{\text{sparse}} = \mathbf{UQ}_{\text{opt}} \tag{31}$$

This final projection ensures that $\mathbf{U}_{\text{sparse}}$ is exactly orthonormal.

Figure 16 shows the basis vector values before and after the sparse rotation, starting from a random initialisation of $\tilde{\mathbf{Q}}$.

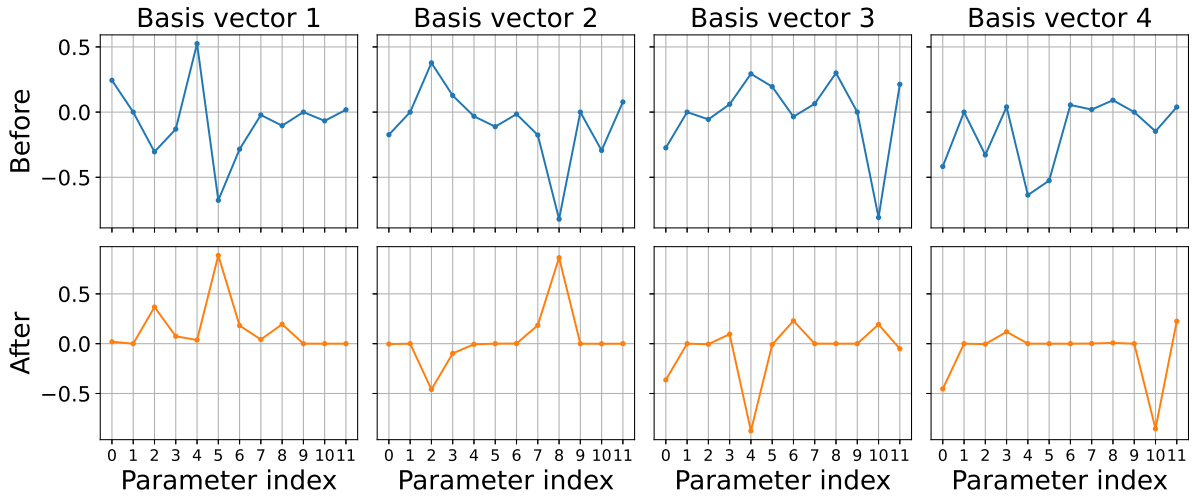


Figure 16: Basis vectors before and after the sparse rotation. Parameter index corresponds to $[w_{osn,r}, w_{osn,b}, w_{osn,g}, w_{osn,y}, w_{b,r}, w_{b,g}, w_{b,y}, w_{r,y}, w_{r,g}, \alpha, W_{g,r}, W_{g,y}]$

While sparsification is visually apparent, Figure 17 provides a more quantitative comparison by showing the L_1 norm before and after the rotation. The reduction in L_1 norm confirms that the rotated basis is sparser than the original SVD basis.

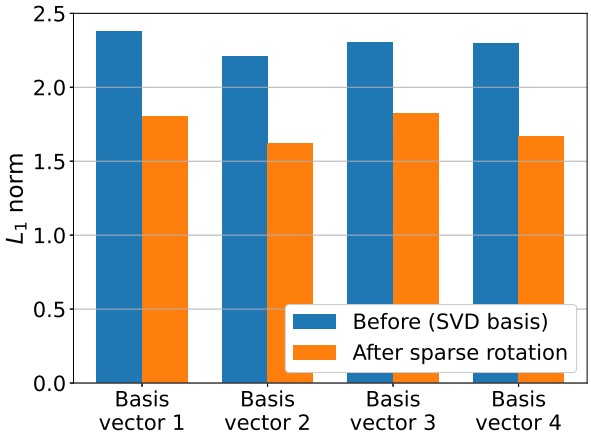


Figure 17: L_1 norm of the basis before and after the sparse rotation.

4.5 HP: Sensitivity to Initialisation

Because the L_1 sparsification objective is non-convex under the orthogonality constraint, this does not guarantee convergence to a global minimum. Different random initialisations of $\tilde{\mathbf{Q}}$ may lead to different final rotations, and therefore different sparse bases for the same underlying subspace. To characterise this variability, we run the optimisation (Section 4.4.3) from 10,000 independent random initialisations and record the final L_1 loss for each. The entire procedure is vectorised across independent runs using `vmap` and compiled with `jit`, enabling efficient parallelisation over multiple random initialisations.

Figure 18 illustrates three example outcomes. Different initialisations produce visibly different rotated bases, confirming that the optimisation landscape contains multiple local minima.

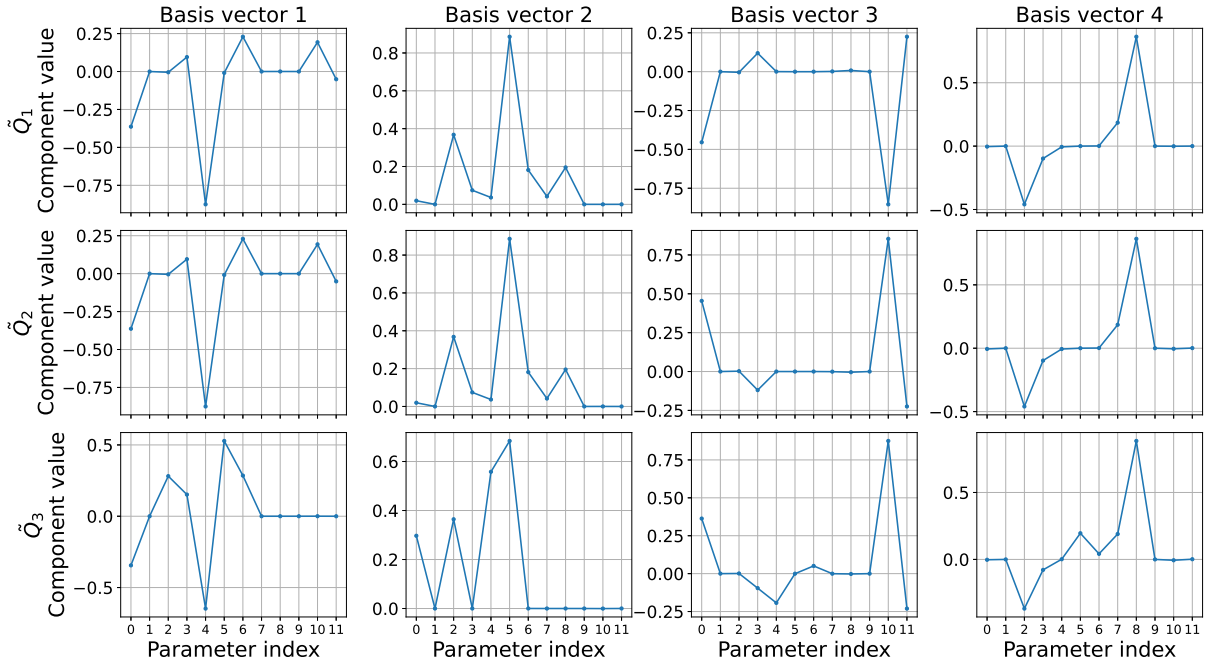
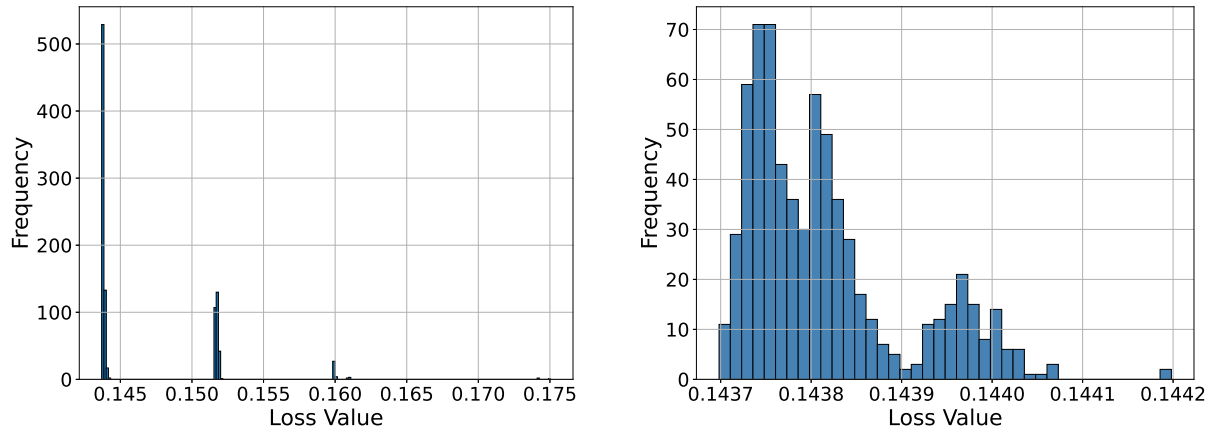


Figure 18: Rotated bases $\mathbf{U}_{\text{sparse}}$ obtained from several different random initialisations of $\tilde{\mathbf{Q}}$. Each row corresponds to one initialisation, columns show the four rotated basis vectors. Differences across rows indicate that the optimiser converges to distinct local minima. Parameter index corresponds to $[w_{osn,r}, w_{osn,b}, w_{osn,g}, w_{osn,y}, w_{b,r}, w_{b,g}, w_{b,y}, w_{r,y}, w_{r,g}, \alpha, W_{g,r}, W_{g,y}]$

Figure 19a shows a histogram of final L_1 losses across all 10,000 runs. The histogram reveals three clusters of typical loss values. We focus on the lowest-loss cluster, which contains the 681 solutions with the smallest value of the sparsity objective. Zooming in on this cluster reveals further structure (Figure 19b). However, the differences in loss between the sub-peaks are small and it is difficult to argue that any one is more significant. Therefore, we treat all 681 solutions as equally valid candidates and consider the full set of sparse bases present in this cluster.



(a) **(b)**

Figure 19: (a) Distribution of final L_1 losses across 10,000 random initialisations (200 bins). Three clusters are visible. (b) A zoom into the lowest-loss cluster.

4.6 HP: Signed Permutation Check

To determine whether the solutions within the lowest-loss cluster represent the same sparse basis or different ones, we compute pairwise dot-product matrices between all retained solutions. For two candidate sparse orthonormal bases $\mathbf{U}^{(i)}_{\text{sparse}}$ and $\mathbf{U}^{(j)}_{\text{sparse}}$, each of shape $[12 \times 4]$, the $[4 \times 4]$ matrix:

$$\mathbf{P}_{ij} = \left(\mathbf{U}^{(i)}_{\text{sparse}}\right)^T \mathbf{U}^{(j)}_{\text{sparse}} \tag{32}$$

measures the alignment between all pairs of basis vectors. If the two bases are related by a signed permutation, then each row and column of $\mathbf{P}_{ij}$ has exactly one entry close to ± 1 and all others close to zero. Figure 20 shows the result of this calculation for 3 sets of bases.

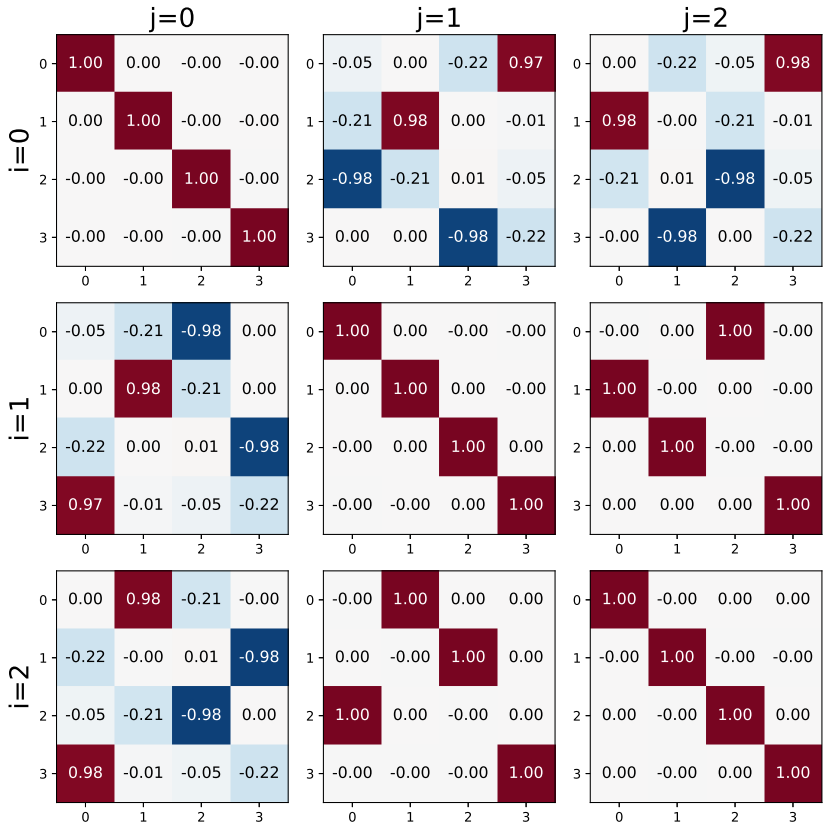


Figure 20: Example dot-product matrices $\mathbf{P}_{ij}$ between pairs of sparse bases. Matrices with a single dominant entry per row and column indicate signed-permutation equivalence.

To quantify this, we define a row residual for each row of P_{ij} as the discrepancy from the ideal signed-permutation pattern:

$$r_i = \sum_{j=1}^{4} |\mathbf{P}_{ij}| - \max_j |\mathbf{P}_{ij}| \tag{33}$$

A row residual of zero indicates that row i contains exactly one entry of magnitude one and all others zero. We then summarise the alignment between two bases by the mean row residual across all four rows, with a small residual indicating signed-permutation equivalence.

Across all pairs within the lowest-loss cluster, we set a row residual threshold of 0.05, below which two bases are considered equivalent up to signed permutation. Only 27.6% of pairs satisfy this threshold. This indicates that the lowest-loss cluster is not a single family of equivalent bases, but rather contains at least two structurally distinct sparse representations of the output-preserving subspace.

4.7 HP: Distinct Sparse Solutions

The signed-permutation analysis reveals that the lowest-loss cluster contains more than one family of sparse bases. We now identify these families explicitly using an iterative grouping process.

We take the first solution in the cluster as a reference matrix $\mathbf{U}_{\text{ref}}^{(A)}$ and compute the mean row residual between $\mathbf{U}_{\text{ref}}^{(A)}$ and every other solution. Solutions with mean row residual below 0.05 are classified as belonging to Group A and the remainder are set aside. Iterating this procedure on each successive remainder group, partitions the full cluster into groups whose bases are equivalent under signed permutation. Table 1 summarises the result.

Iteration	Group	Solutions	Remainder
1	A	172	509
2	B	142	367
3	C	259	108
4	D	108	0
	Total	**681**	

Table 1: Iterative partitioning of the lowest-loss cluster into signed-permutation equivalence classes.

The empty remainder after iteration 4 confirms that every solution belongs to exactly one of the four sub-families. The lowest-loss cluster therefore contains four distinct sparse bases, A through D, each internally consistent up to signed permutation.

4.8 HP: Sparse Bases Interpretation

Having identified four distinct families of sparse bases, we select one representative from each group and examine the structure of the resulting basis vectors.

Figure 21 shows the four representative bases overlaid. Each basis vector is plotted over the twelve parameters of θ, with component value indicating the magnitude of the corresponding weight entry.

Each basis vector identifies a specific combination of synaptic weights that can change together without altering mitral cell output. A sparse basis vector dominated by a single parameter would indicate that the corresponding synapse can vary in isolation. A denser vector, as observed across all four representative bases shown here, would indicate that a coordinated co-variation of several weights is required to maintain the same output.

A key finding of this analysis is that the sparse bases are not arbitrary. All 681 solutions in the lowest-loss cluster (Figure 19b) span the same four-dimensional output-preserving subspace, but they express that subspace using different sparse basis vectors. Rather than converging to a single sparse basis, the solutions fall into four distinct families (Groups A, B, C, and D), each equivalent up to signed permutation internally but structurally distinct across families (Figure 21). This pattern is important because it shows that optimisation does not recover one unique sparse basis, but repeatedly returns a small set of alternative sparse coordinate systems for the same subspace. This reproducibility shows that the

result is structured rather than random. The sparsity is produced by optimisation, as intended, but it is not an artefact of a single lucky initialisation, the same few families reappear across thousands of independent runs.

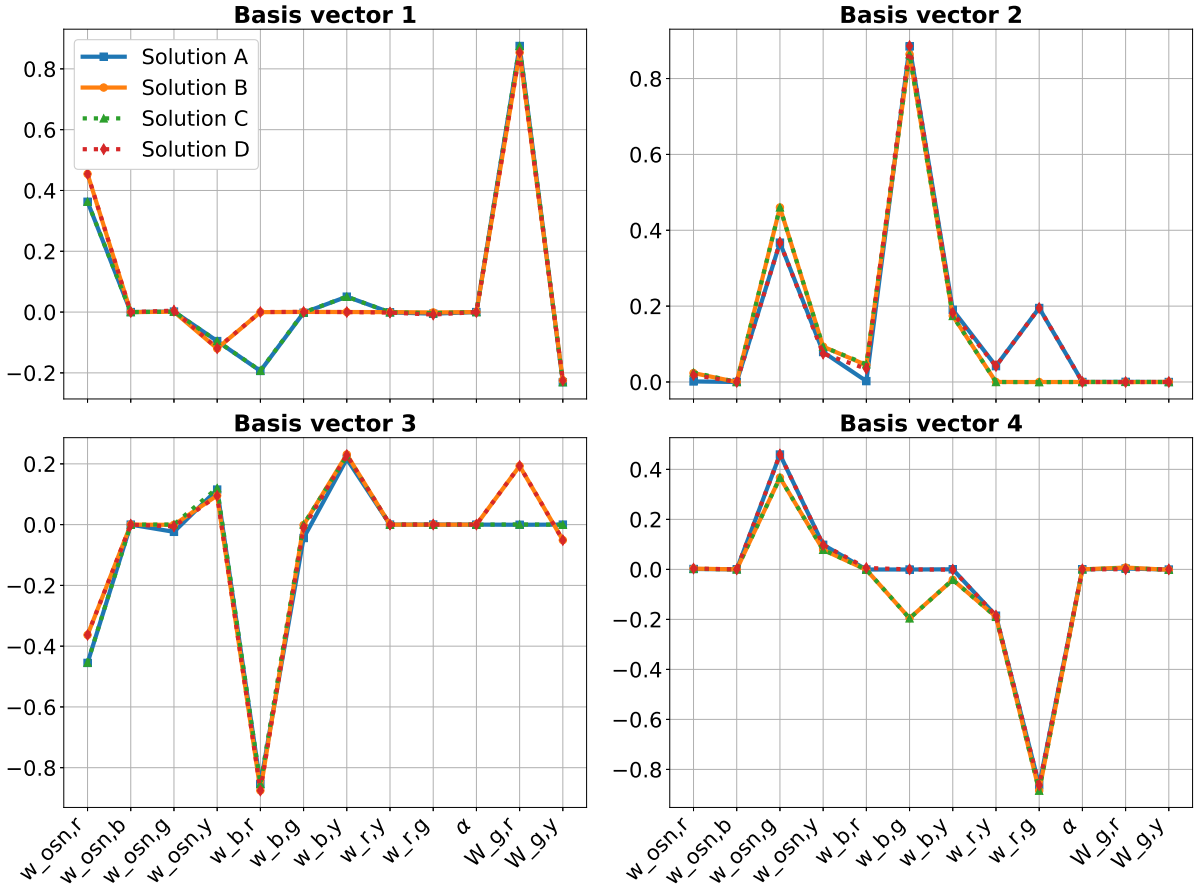


Figure 21: Representative sparse bases for Groups A, B, C, and D overlaid. Each panel corresponds to one of the four basis vectors. The four groups present structurally different sparse directions within the same four-dimensional subspace.

4.9 HP: Verification of Output-Preserving Directions

The sparse bases U_A, U_B, U_C and U_D identified in Section 4.8 represent candidate directions along which synaptic weights could vary with little effect on mitral cell output. They were obtained by requiring the quadratic term in Equation (20) to vanish. However, this condition alone is insufficient to guarantee constant output. Perturbations along each sparse direction must also be orthogonal to the gradient, i.e. $\text{grad}^\top \Delta\theta \approx 0$. Only if both conditions hold can the sparse basis vectors be regarded as output-preserving directions.

4.9.1 Orthogonality to the Gradient

To verify whether the sparse bases are orthogonal to the gradient, we evaluate the gradient at each converged baseline θ_o from Section 3.5. Since the sparse bases u_k were identified as a shared structure across all 570 retained baselines (Section 4.8), orthogonality can be tested at every θ_o. For each unit basis vector $\hat{u}_k$ (with $\|\hat{u}_k\| = 1$), we form the normalised gradient $\hat{g}_i = \text{grad}_i / \|\text{grad}_i\|$ and record the dot product $\hat{u}_k \cdot \hat{g}_i$. A value close to zero indicates that the corresponding sparse direction is approximately orthogonal to the

gradient at that baseline, so that the linear term in Equation (20) is negligible. Figure 22 shows the distribution of these dot products across all 570 baselines and all four basis vectors.

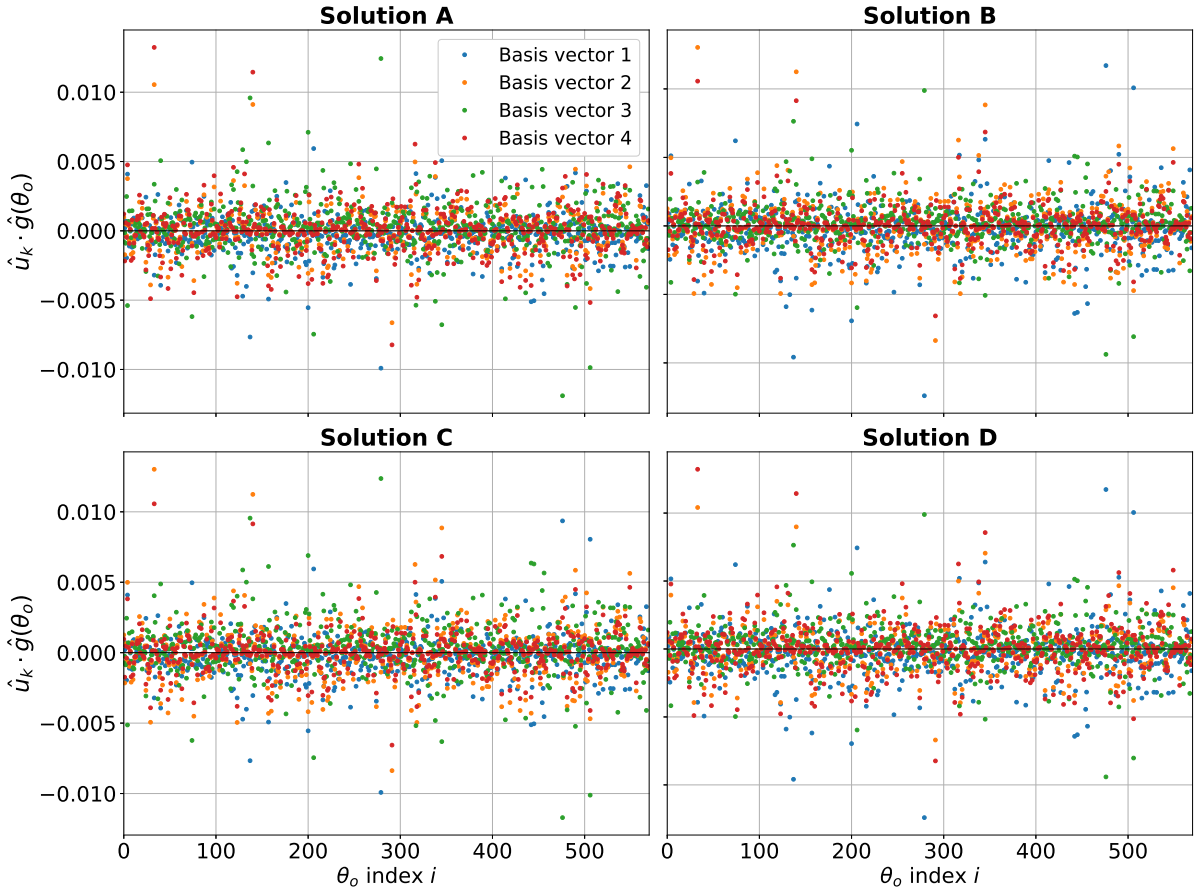


Figure 22: Normalised dot products $\hat{u}_k \cdot \hat{\mathrm{g}}(\theta_o)$ between each sparse basis vector $\hat{u}_k$ (from U_A, U_B, U_C, or U_D) and the output gradient at every retained baseline θ_o. Values cluster near zero, confirming that the shared sparse directions are typically almost orthogonal to the gradient.

Across all four sparse families, the mean $|\hat{u}_k \cdot \hat{\mathrm{g}}|$ over retained baselines is $\mathcal{O}(10^{-4})$, with a maximum of $\sim 10^{-2}$ for a small number of (i, k) pairs. Geometrically, $|\hat{u}_k \cdot \hat{\mathrm{g}}| = 0.01$ corresponds to an angle within $0.6°$ of orthogonality. However, for a step in $\Delta\theta = \varepsilon \hat{u}_k$, the linear contribution may not be non-negligible at large step sizes. We therefore proceed to directly verify output preservation by stepping along the sparse directions and measuring the resulting change in y.

4.9.2 Stepping Along Sparse Bases

For each of the retained baseline parameter sets θ_o, (570 initialisations) we construct perturbed parameters by stepping along a unit direction $\hat{u}_k$, a column of one of the sparse basis matrices U_A, U_B, U_C or U_D, at a range of step sizes ε:

$$\theta(\varepsilon) = \theta_o + \varepsilon\, \hat{u}_k \qquad \|\hat{u}_k\| = 1 \tag{34}$$

For each perturbed parameter set, we solve the circuit fixed-point equations to obtain the new steady-state mitral cell output, and measure the absolute change relative to the unperturbed value y^*:

$$\Delta y(\varepsilon) = |y(\theta(\varepsilon)) - y^*| \tag{35}$$

To produce a step size that is interpretable across different parameter configurations, we normalise ε by the relative magnitude of the perturbation:

$$\tilde{\varepsilon} = \varepsilon \left\| \frac{\Delta\theta}{\theta} \right\| \tag{36}$$

This normalised step size measures the perturbation as a fraction of the current parameter values, making it directly comparable across θ's with different scales. For each direction and step size, results are averaged over all 570 retained baseline solutions.

Figure 23 shows the relative change in mitral cell output $\Delta y(\varepsilon)/y*$, where y^* is the unperturbed steady-state output, as a function of normalised step size $\tilde{\varepsilon}$, for each of the four sparse directions from U_A, U_B, U_C and U_D .

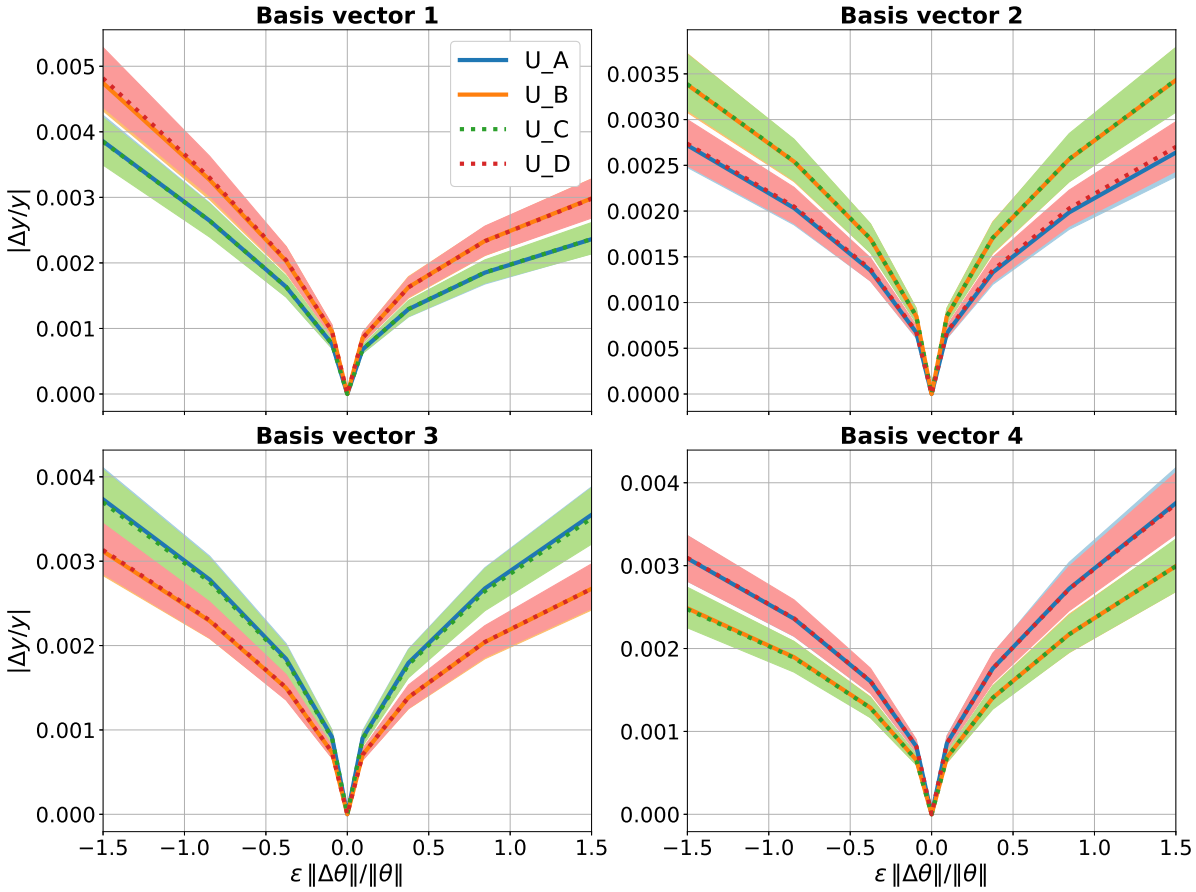


Figure 23: Relative change in mitral cell output $\Delta y/y^*$ as a function of normalised step size, for each of the four sparse directions from U_A, U_B, U_C and U_D. Each panel corresponds to one basis vector. The solid line shows the mean across baseline solutions, while the shaded region represents the confidence interval of 95%(CI).

Across all four basis vectors and all four sparse bases sets (A, B, C, D), mitral output y stays close to its baseline value over a wide range of step sizes $\tilde{\varepsilon}$. Even for parameter perturbations on the order of 100% relative to $\|\theta\|$, the relative change in output remains

below 1%. The narrow confidence intervals show that this insensitivity is consistent across the 570 baseline solutions.

4.9.3 Comparison with Random Perturbations

To place these results in context, Figure 23 repeats the same experiment but replaces the sparse directions with random twelve-dimensional unit vectors.

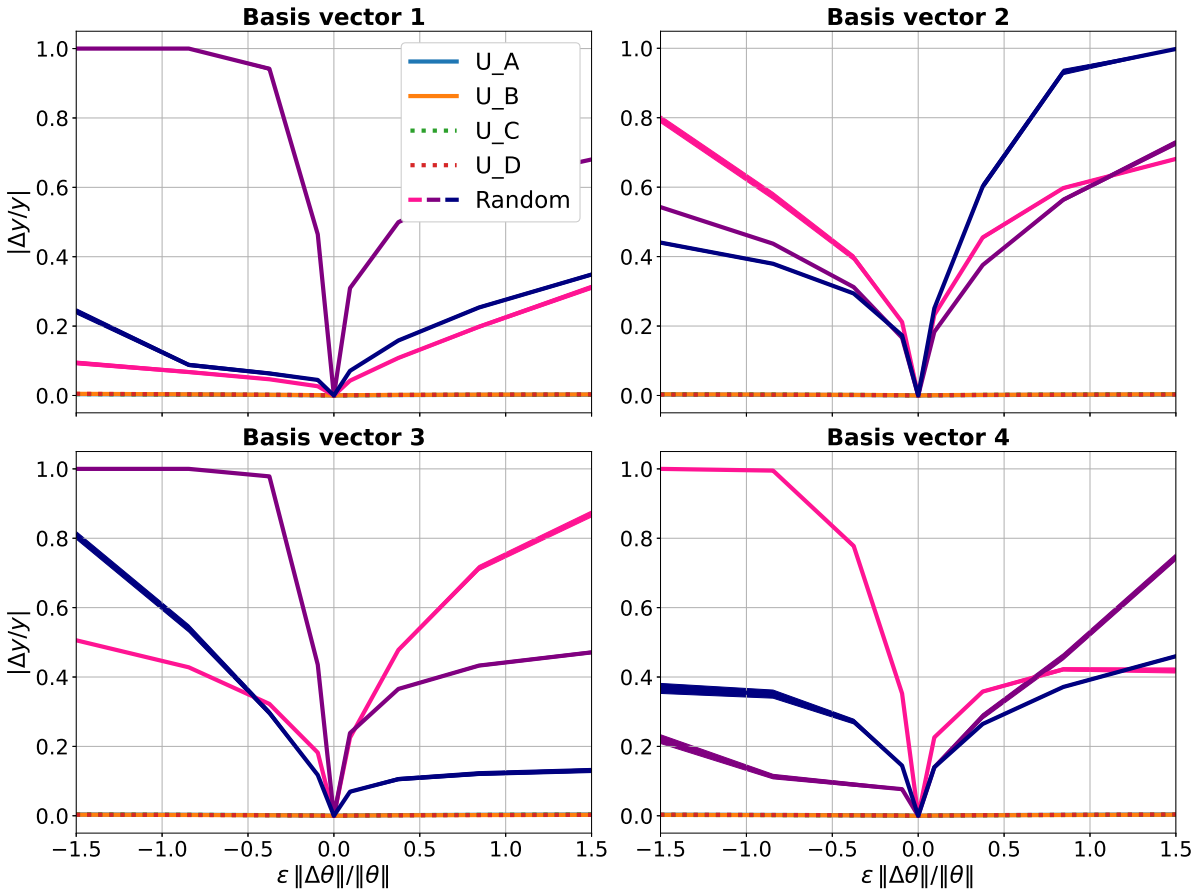


Figure 24: Figure 23, with the addition of pink, purple, dark blue curves showing the output change when stepping along 3 random directions.

The contrast is striking. Steps along random directions produce output changes that grow rapidly with step size. On the other hand, steps along the sparse basis directions produce changes that are substantially smaller at the same normalised step size. This confirms that the sparse bases correspond to directions along which the output is insensitive to.

These results confirm that the small but non-zero gradient dot products ($\mathcal{O}(10^{-2})$ to $\mathcal{O}(10^{-4})$) are sufficiently close to zero to preserve output in practice. Together, the orthogonality check and the stepping measurements validate the sparse basis directions as biologically plausible candidates for homeostatic plasticity. A circuit recovering from sensory deprivation could traverse these directions, adjusting synaptic weights according to the specific combinations encoded in U_A, U_B, U_C, or U_D, while maintaining stable mitral cell output throughout. The sparsity of these basis vectors implies that only a small subset of synaptic connections need be modified simultaneously, offering an experimentally testable account of how output homeostasis might be implemented in the olfactory bulb.

5 Biological Constraints

Experimental recordings show that green cells receive more frequent and stronger inhibitory signals from blue cells following sensory deprivation, corresponding to an increase in $w_{b,g}$ (E. Galliano, personal communication, 2026, Appendix B(A)). Inspection of the sparse bases in Figure 21 reveals that Basis vector 2 contains a dominant $w_{b,g}$ component, identifying it as the direction most consistent with this experimental observation. Importantly, the basis vector also specifies the compensatory adjustments that must accompany this change if mitral cell output is to remain constant. In particular, the model predicts that the increase in $w_{b,g}$ should be accompanied by corresponding increases in $w_{osn,g}$ and $w_{r,g}$, representing stronger synaptic drive from OSNs and ETC (red) cells onto the axonic interneuron population. This is biologically plausible and consistent with classical homeostatic plasticity, as sensory deprivation leads to an upscaling of excitatory synapses in an attempt to compensate for reduced input.[18]

However, these results are incomplete as it has also been observed that following sensory deprivation, the axon initial segment (AIS) in axonic (green) interneurons shrinks.[8] The AIS is the cellular subcompartment where voltage-gated sodium channels are clustered and so where action potentials are initiated. Therefore, a shorter AIS is associated with decreased neuronal excitability. A decrease in neural excitability has also been observed experimentally by measuring the input–output relationship between current injection amplitude and firing rate.[8] In the model, these observed changes require variations in the Hill exponent n_g. This effect was not captured in the original analysis, in which n_g was held fixed, potentially leading to compensatory changes being absorbed by other parameters. To address this, we extend the parameter vector to include n_g as a free variable and repeat the full analysis. The question we now ask is: given that n_g decreases and $w_{b,g}$ increases (while keeping n_r, n_b, and n_y constant[8]), what compensatory adjustments in the remaining parameters are required to preserve mitral cell output?

5.1 Why a Single Operating Point Cannot Reveal n_g Sensitivity

Treating n_g as a free parameter requires care, because the choice of operating point determines whether n_g has any effect on the circuit output at all. In all prior analyses, the circuit was constrained to operate at $x = 1$, with a target firing rate of 0.5 for all populations. This operating point was chosen because the semi-saturation constant of each Naka–Rushton unit was normalised to unity (Section 3.2), so that at $x = 1$:

$$\sigma(1) = \frac{1^n}{1^n + 1^n} = \frac{1}{2} \tag{37}$$

independently of the Hill exponent n. As a result, at this operating point the green cell output g is insensitive to n_g, and since g propagates through the circuit to influence r and y (Equations (7)-(9)), all downstream firing rates are equally insensitive. Consequently, the gradient of the mitral cell output with respect to n_g vanishes:

$$\left.\frac{\partial y}{\partial n_g}\right|_{x=1} = 0 \tag{38}$$

and since the gradient is zero at $x = 1$, the corresponding row and column of the Hessian $\mathbf{H}$ are also zero. No information about compensatory changes involving n_g can therefore

be extracted from a single-operating-point analysis because the Taylor expansion of y is blind to n_g by construction, regardless of the parameter set θ.

5.2 Optimisation Using Two Fixed Points

To break this degeneracy, we introduce a second operating point. Shifting the single operating point from $x = 1$ to some other value would, in principle, introduce sensitivity to n_g. However, we still require this operating point to anchor the circuit at the half-max reference state (all $r, g, b, y = 0.5$ at $x = 1$). We therefore augment the existing constraints by requiring the circuit to satisfy two simultaneous conditions:

- **Operating point 1** ($x = 1.0$): all four populations r, g, b, y are constrained to fire at 0.5, as before. This preserves the dynamic-range condition that anchors the baseline parameterisation.
- **Operating point 2** ($x = 0.5$): only the mitral cell y is constrained to achieve a target output of 0.2. At this input level, $\sigma(0.5)$ depends explicitly on n_g, breaking the degeneracy and allowing the Taylor expansion to carry information about compensatory changes involving n_g.

This two-point formulation is a minimal extension that restores sensitivity to n_g while preserving continuity with all prior analyses. The parameter vector is extended to $\theta \in \mathbb{R}^{13}$, with n_g appended as the thirteenth element. We seek initial parameter sets that simultaneously satisfy both operating points, using the combined loss function:

$$L(\theta) = \sum_{i=1}^{m} (y_i - 0.2)^2 \Big|_{x=0.5} + \sum_{i=1}^{m} \left[(r_i - 0.5)^2 + (g_i - 0.5)^2 + (b_i - 0.5)^2 + (y_i - 0.5)^2 \right] \Big|_{x=1} \tag{39}$$

where $m = 2$ is the number of glomeruli. The first term enforces the output target at the second operating point, where the circuit is sensitive to n_g, while the second term preserves the dynamic-range conditions established in Section 3.5. This is minimised using the Adam optimiser from 1000 random initialisations, with all weights constrained to be non-negative and $\alpha \in [0, 1]$, as before.

As shown in Figure 25, not all initialisations converge to valid solutions. Of the 1,000 initialisations, only 214 converge to solutions satisfying both target conditions within tolerance (Loss $< 10^{-6}$), a substantially lower success rate than in the single operating point case. This is expected, since requiring the model to satisfy two operating points simultaneously imposes a stronger constraint. These 214 retained solutions form the baseline parameter set for the constrained direction analysis that follows.

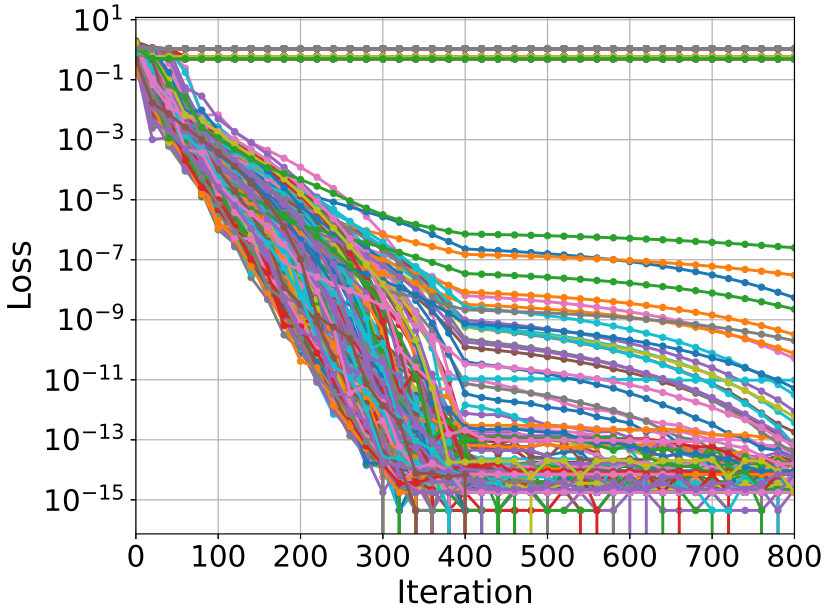
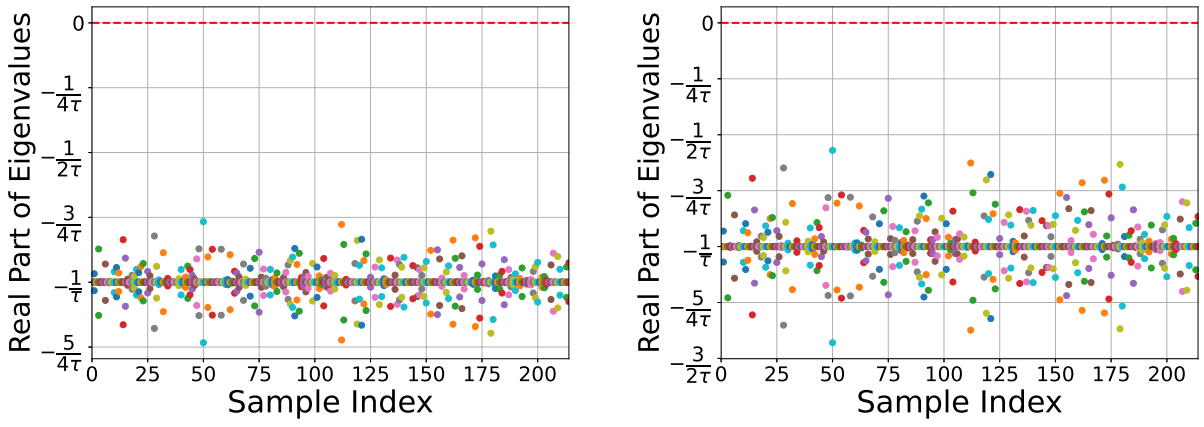


Figure 25: Loss convergence during two-operating-point optimisation. Valid solutions converge reliably to near-zero loss.

5.3 Stability Verification of Two-Operating-Point Solutions

The 214 optimised two-operating-point solutions are only biologically plausible if they correspond to stable fixed points of the network dynamics. As in Section 3.6, local stability is assessed by linearising the network dynamics around each fixed point and examining the eigenvalues of the resulting system matrix $\mathbf{A}$. A fixed point is locally stable if and only if all eigenvalues have strictly negative real part, $\mathrm{Re}(\lambda_i) < 0$ for all i. Since the two-operating-point solutions are required to satisfy two simultaneous constraints, stability must be verified independently at both operating points: at $x = 1.0$, where the target mitral output is $y = 0.5$, and at $x = 0.5$, where the target mitral output is $y = 0.2$.



(a) **(b)**

Figure 26: Real part of the eigenvalues of $\mathbf{A}$ for each of the 214 two-operating-point parameter sets, evaluated at (a) operating point $x = 1.0$, target mitral output $y = 0.5$, and (b) operating point $x = 0.5$, target mitral output $y = 0.2$.

Figure 26 shows the real parts of the eigenvalues of $\mathbf{A}$ across all 214 parameter sets at both operating points. In every case, all eigenvalues have strictly negative real parts, confirming that all 214 two-operating-point solutions are locally stable. Every eigenvalue lies well below the stability boundary marked by the dashed red line at zero, providing confidence that these solutions represent stable operating points of the circuit.

5.4 Homeostatic Plasticity: Varying n_g

We repeat the Taylor expansion analysis from Section 4.1, now applied to the extended thirteen-dimensional parameter vector. As before, we seek directions that are approximately orthogonal to the gradient and lie in the near-null subspace of the Hessian $\mathbf{H}$. We compute the Hessian of mitral output at the low-input operating point $x = 0.5$, where n_g is non-degenerate and affects y.

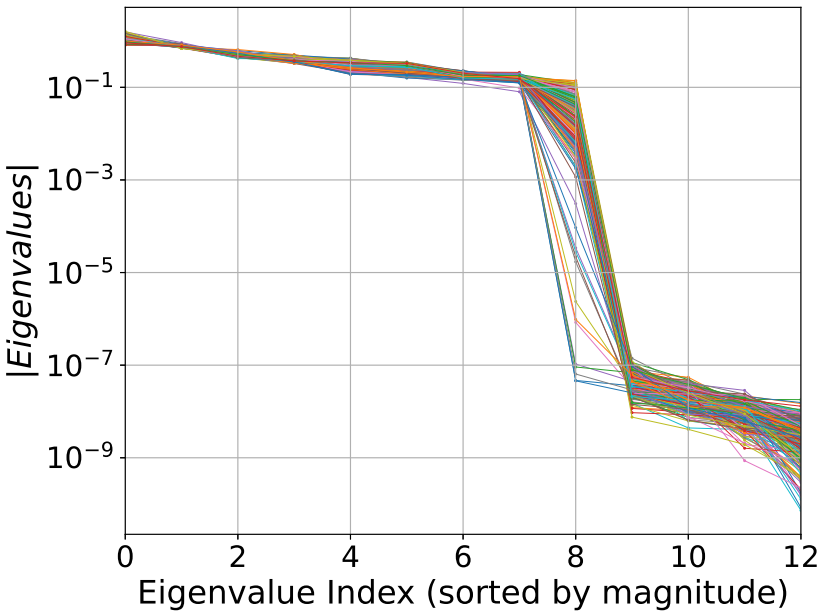


Figure 27: Distribution of eigenvalue magnitudes of the Hessian $\mathbf{H}$ across 214 parameter initialisations.

For each optimised parameter θ_o, we compute the Hessian $\mathbf{H_i} = \nabla^2_\theta y(\theta_o)$ and examine its eigenvalues. Figure 27 shows the distribution of Hessian eigenvalue magnitudes across all 214 retained parameter sets. Similarly to the previous twelve-parameter analysis, four consistently small eigenvalues emerge across initialisations.

We stack the eigenvectors corresponding to eigenvalues smaller than 10^{-6} across all retained initialisations and apply singular value decomposition to identify directions that reoccur most frequently across eigenvectors. Figure 28 shows the resulting singular values, where a clear gap identifies the four dominant directions that recur consistently across parameter sets.

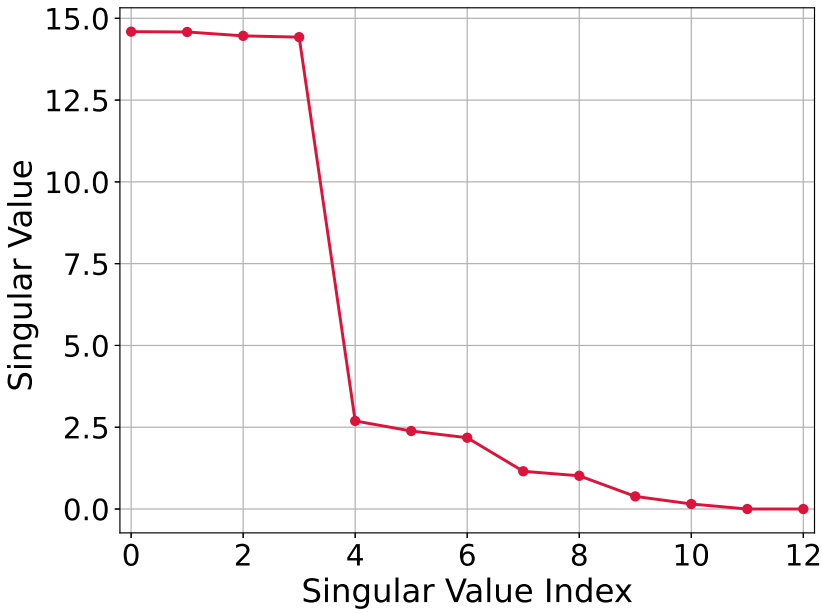


Figure 28: Singular values of the matrix of stacked near-null eigenvectors across all parameter sets

We next apply the sparse-rotation procedure of Section 4.4, minimising $\|\hat{\mathbf{U}}\|_1$ for $\hat{\mathbf{U}} = \mathbf{U}\mathbf{Q}$ over many random orthogonal initialisations $\tilde{\mathbf{Q}}$. A histogram of the final losses separates a dominant low-loss cluster (data not shown explicitly). Within the lowest cluster, a signed-permutation check (Section 4.6) reveals a single equivalence group, i.e only one distinct four-dimensional sparse basis. Figure 29 shows the four resulting basis vectors. In contrast to the twelve-parameter case, which produced four distinct sparse basis families, the sparsification here produces a single equivalence group.

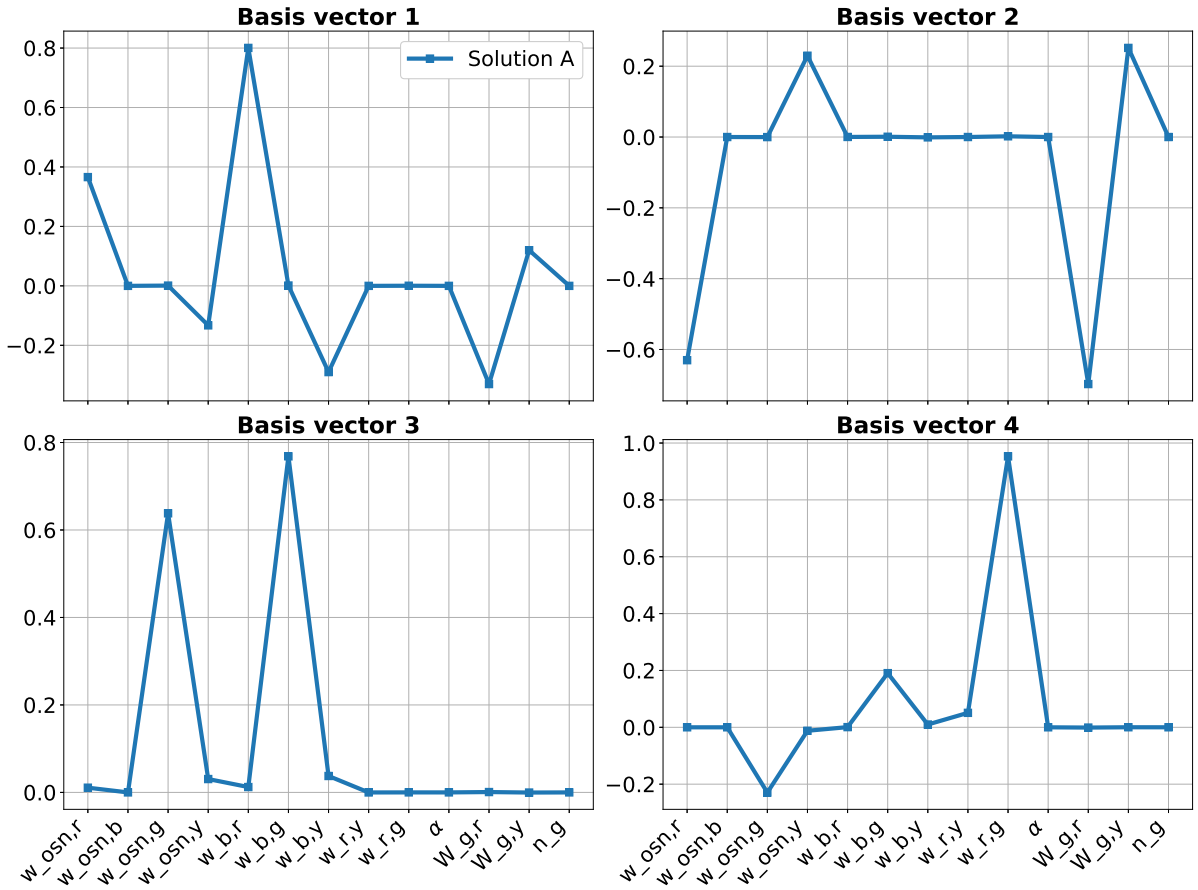


Figure 29: Sparse bases for Group A, where each panel corresponds to one of the four basis vectors.

Inspection of the basis vectors in Figure 29 reveals that the n_g component is zero in all four directions, meaning that any step along a basis vector leaves n_g unchanged. This contradicts the experimental observation that n_g decreases during sensory deprivation, and indicates that the near-null subspace of the Hessian alone is insufficient to capture the full compensatory change. We therefore adopt a different approach. Rather than searching for directions of least sensitivity, we instead impose a non-zero change in n_g and determine the remaining components of $\Delta\theta$ so that the circuit output is preserved as much as possible.

5.5 Constrained Direction Analysis

We now seek parameter perturbations $\Delta\theta$ that are consistent with two experimental observations: a decrease in n_g and an increase in $w_{b,g}$. Rather than identifying directions of minimal output sensitivity as before, we pin n_g to change, and ask what compensatory adjustments in the remaining 12 parameters are required to keep the circuit output unchanged.

Let $v \in \mathbb{R}^{13}$ denote the unit vector in the n_g direction. Any parameter perturbation $\Delta\theta = s \in \mathbb{R}^{13}$ can be decomposed into components parallel and orthogonal to v:

$$s = s_{\parallel} + s_{\perp} = v(v^T s) + (I - vv^T)s \tag{40}$$

Since the change in n_g is prescribed by the experimental observation, we fix $v^T s = 1$, representing a unit step in n_g, and allow only the orthogonal component $s_{\perp} = (I - vv^T)s$

to be optimised. The choice $v^T s = 1$ sets the scale of the n_g perturbation and the resulting compensatory directions $s_\perp^*$ scale proportionally, so the relative structure of the solution is independent of this choice. The optimisation seeks the twelve weight adjustments that minimise the quadratic term in Equation (20):

$$\min_{s_\perp} \; s^T \mathbf{H}(\theta_\mathbf{o}) s \quad \text{subject to} \quad v^T s = 1 \tag{41}$$

where $\mathbf{H}(\theta_o)$ is the Hessian of the mitral output evaluated at $x = 0.5$, the operating point at which n_g measurably affects y, and θ_o is a baseline from the set of valid two-operating-point parameter initialisations.

Since $v = [0,\, 0,\, \ldots,\, 0,\, 1]^T$ has its nonzero entry in the n_g position, the outer product vv^T and the projection $I - vv^T$ take the block forms:

$$vv^T = \begin{bmatrix} \mathbf{0}_{12\times 12} & \mathbf{0}_{12\times 1} \\ \mathbf{0}_{1\times 12} & 1 \end{bmatrix} \qquad I - vv^T = \begin{bmatrix} I_{12} & \mathbf{0}_{12\times 1} \\ \mathbf{0}_{1\times 12} & 0 \end{bmatrix} \tag{42}$$

Partitioning the symmetric matrix $\mathbf{H}$ similarly:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H_1} & \mathbf{h_2} \\ \mathbf{h_2^T} & h_3 \end{bmatrix} \tag{43}$$

where $\mathbf{H}_1 \in \mathbb{R}^{12\times 12}$, $\mathbf{h}_2 \in \mathbb{R}^{12\times 1}$, and $h_3 \in \mathbb{R}$. Substituting $s = [s_\perp^T,\; 1]^T$ gives:

$$s^T \mathbf{H} s = \begin{bmatrix} s_\perp \\ 1 \end{bmatrix}^T \begin{bmatrix} \mathbf{H_1} & \mathbf{h_2} \\ \mathbf{h_2^T} & h_3 \end{bmatrix} \begin{bmatrix} s_\perp \\ 1 \end{bmatrix} = s_\perp^T \mathbf{H}_1 s_\perp + 2\mathbf{h}_2^T s_\perp + h_3 \tag{44}$$

Since h_3 is constant with respect to $s_\perp$, minimising over $s_\perp$ is achieved by differentiating the remaining terms and setting the result to zero:

$$\frac{\partial}{\partial s_\perp}\left(s_\perp^T \mathbf{H_1} \mathbf{s}_\perp + \mathbf{2h_2^T} \mathbf{s}_\perp\right) = 2\mathbf{H_1} \mathbf{s}_\perp + \mathbf{2h_2} = \mathbf{0} \tag{45}$$

The full optimal constrained perturbation that minimises $s^T \mathbf{H} s$ is therefore:

$$s^* = v + s_\perp^* \qquad s_\perp^* = -\mathbf{H_1^{-1}} \mathbf{h_2} \tag{46}$$

5.5.1 Numerical solution: pseudoinverse

The constrained optimisation problem is solved analytically for each of the 214 baseline initialisations, producing a set of candidate compensatory directions. However, the solution requires inverting $\mathbf{H_1}$ (Equation (46)), which is singular by construction, i.e. it maps at least one non-zero vector to zero, so a standard matrix inverse does not exist.

Inverting a matrix amplifies each direction in proportion to $1/\sigma$, where σ is the corresponding singular value. For directions along which $\sigma \approx 0$, a naive inversion would amplify these directions by an arbitrarily large factor, producing an unreliably large $s_\perp^*$. Therefore, we avoid this by using the pseudoinverse $\mathbf{H_1}^+$, computed with a regularisation threshold of `rcond`$=10^{-6}$, which treats any singular value smaller than $10^{-6}\sigma_{max}$ as exactly zero, contributing nothing to the solution along those directions.

$$s_\perp^* = -\mathbf{H_1}^+ \mathbf{h_2} \tag{47}$$

Intuitively, the pseudoinverse adjusts only the directions in which the output is sensitive to parameter changes, and leaves the insensitive directions untouched. Since we are looking for $s^*_\perp$, the compensatory adjustment forced by the pinned step in n_g, there is no reason to move along directions that do not affect the output. Therefore, this produces the smallest possible compensatory perturbation consistent with the imposed constraint.

5.5.2 Average Direction Across Solutions

For each of the 214 baseline parameter sets θ_o, we compute $\mathbf{H}(\theta_o)$ and solve for $s^*_{\perp,k}$, then reconstruct the full candidate direction by combining the pinned n_g component v with the optimised orthogonal component:

$$s^*_k = v + s^*_{\perp,k} \tag{48}$$

Repeating this across all 214 baselines produces a collection of candidate compensatory directions, stored row-wise in a matrix, where each row corresponds to one direction s^*_k. We normalise each direction to unit length before further analysis, ensuring that the subsequent comparison across baselines reflects differences in direction rather than differences in scale:

$$\hat{r}_k = \frac{s^*_k}{\|s^*_k\|_2} \tag{49}$$

Figure 30 shows the normalised directions $\hat{r}_k$ across all 214 baselines, showing how each parameter must adjust in response to an increase in n_g. Note that after normalisation the n_g component is no longer exactly unity, but the relative pattern of parameter covariation is preserved. Since the experimental observations correspond to a decrease in n_g rather than an increase, we flip the sign of all directions before applying the experimental constraints.

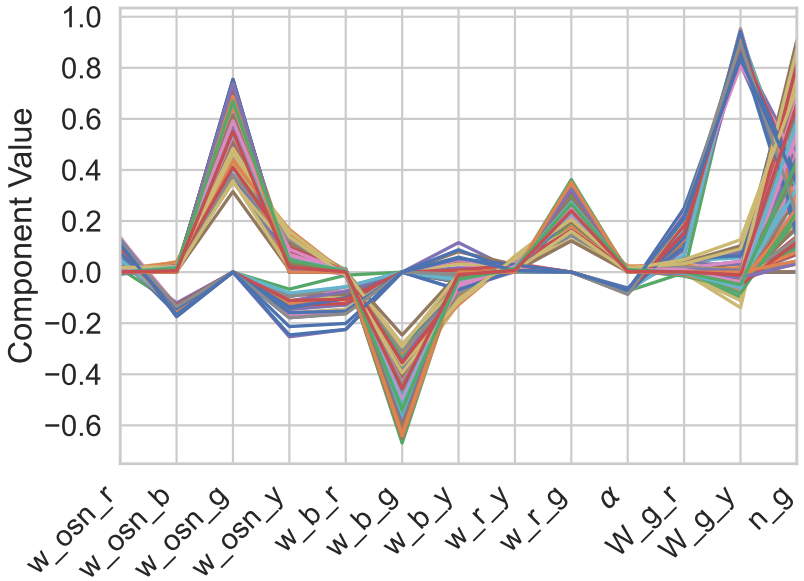


Figure 30: Normalised constrained step directions $\hat{r}_k$ at each of the 214 baseline parameters $\theta^\star_k$.

The normalised directions $\hat{r}_k$ fall into two distinct families, visible as two clusters in Figure 30. Only those directions that simultaneously exhibit an increase in $w_{b,g}$ and a decrease in n_g are retained. Among the 187 retained directions (Figure 31), all show consistent patterns of parameter co-variation, differing only in the magnitude of individual components.

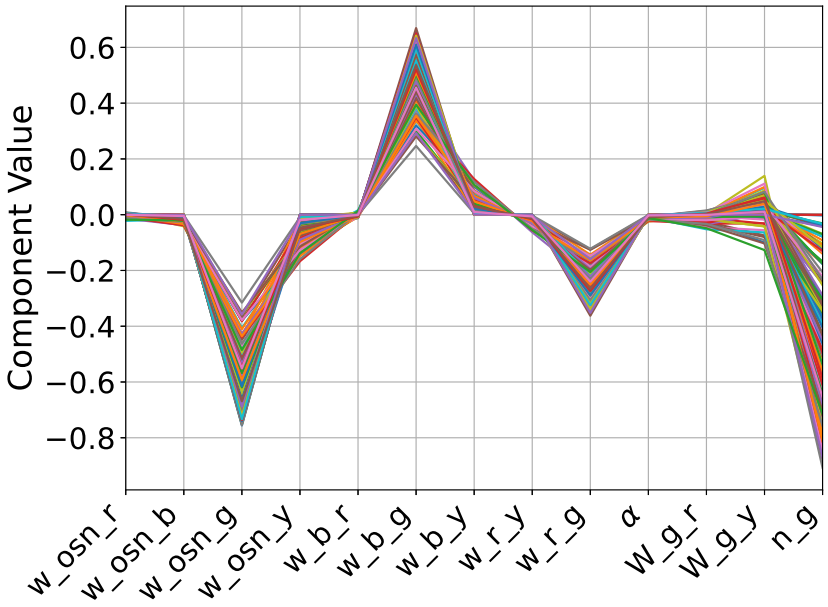


Figure 31: Candidate constrained directions $\hat{r}_k$ satisfying decreasing n_g and increasing $w_{b,g}$ (sign-flipped from Figure 30). 187 of 214 baseline solutions retained.

Requiring the quadratic term in Equation (20) to be approximately zero is not sufficient for output preservation. We must also verify that the linear term vanishes, i.e. that $\text{grad}^\top s_k^* \approx 0$, where the gradient is evaluated at the same baseline θ_o used to compute s_k^*.

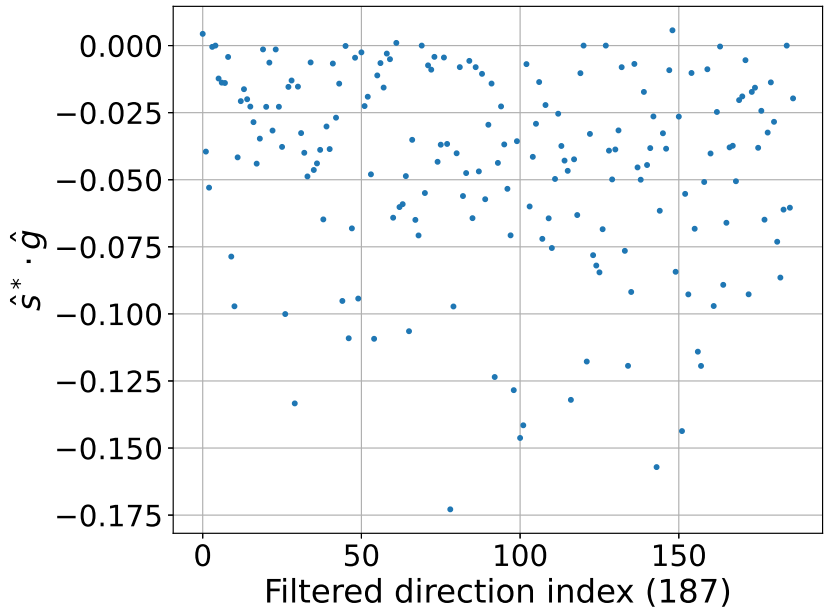


Figure 32: Normalised dot products $\hat{s}_k^* \cdot \hat{g}$ between each retained candidate direction $\hat{s}_k^*$ and the output gradient at the corresponding baseline θ_o. Since both vectors are normalised, the values are equivalent to the cosine of the angle between the two vectors.

Figure 32 shows that $| \hat{s}^* \cdot \hat{g} | < 0.175$, where $\hat{s}^*$ and $\hat{g}$ are the normalised grad and s_k^* vectors. This value correspond to an angular deviation of $< 10°$ from exact orthogonality to the gradient, which while not negligible, is small.

Since all 187 retained directions share a consistent pattern of parameter co-variation, we summarise them by taking their average, producing a single representative direction (Figure 33).

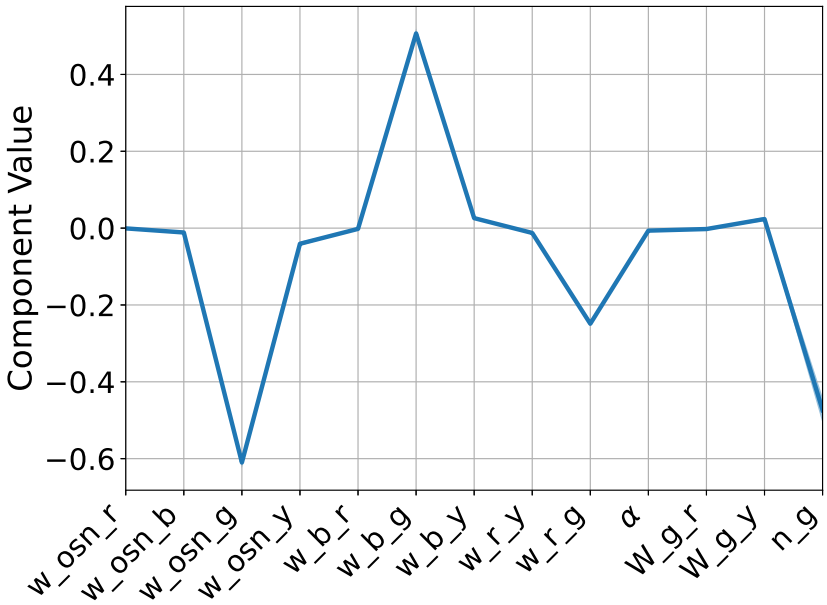


Figure 33: Average of the 187 retained directions $\hat{r}_k$.

5.5.3 Stepping Along Constrained Directions

By construction, we know that the average direction (Figure 33) does not lie in the null space of $\mathbf{H}$ and therefore it may be approximately, but not exactly, output-preserving. To quantify the deviation, we step along it in parameter space and measure the resulting change in mitral cell output. For a normalised direction $\Delta\theta$, we perturb each baseline θ_o as:

$$\theta(\varepsilon) = \theta_o + \varepsilon \, \Delta\theta \tag{50}$$

For each perturbation, we re-solve the circuit fixed point and evaluate the resulting mitral cell output $y(\varepsilon)$, varying ε over the range $[-2, 2]$. To ensure comparability across baseline parameter sets of different scales, each step is additionally expressed as a fraction of the baseline parameter norm:

$$\tilde{\varepsilon} = \frac{\varepsilon \, \|\Delta\theta\|}{\|\theta^*\|} \tag{51}$$

The corresponding relative change in mitral output is then defined as:

$$\left| \frac{\Delta y}{y} \right| = \left| \frac{y(\varepsilon) - y(0)}{y(0)} \right| \tag{52}$$

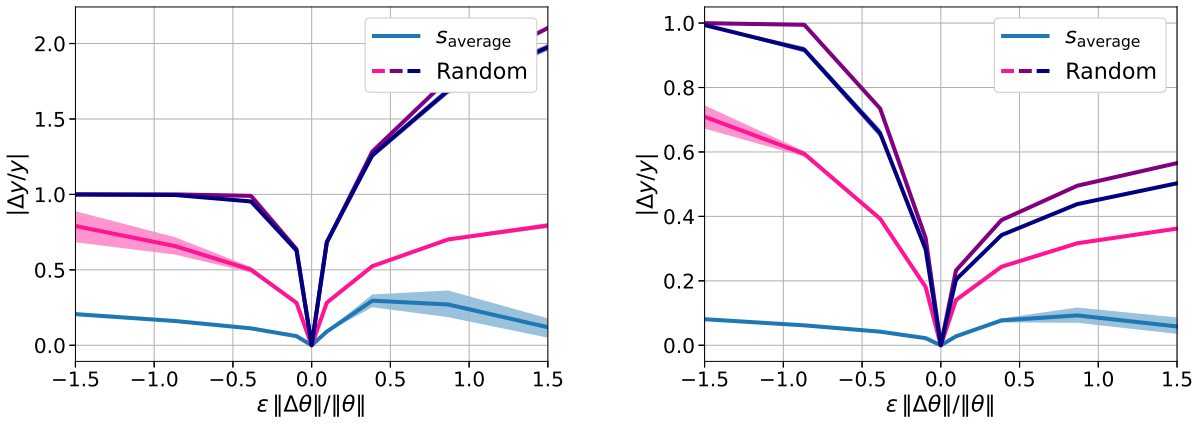


Figure 34: Average relative change in mitral cell output as a function of normalised step size $\varepsilon\|\Delta\theta\|/\|\theta_o\|$ along the average direction $\hat{r}_k$, for $\varepsilon \in [-2, 2]$, at two values of sensory input x. The solid line shows the mean over all retained baseline solutions and the shaded bands denote 95% confidence intervals (CI). Input: (a) $x = 0.5$, (b) $x = 1$.

Figure 34 shows the relative output change when stepping along the n_g pinned average direction at $x = 0.5$ and $x = 1$. The response is noticeably asymmetric in $\tilde{\varepsilon}$, i.e. for postive steps, $|\Delta y/y|$ rises more steeply than for negative steps of equal magnitude. In some directions it exhibits a non-monotonic pattern, first increasing and then decreasing as $\tilde{\varepsilon}$ becomes more positive. This asymmetry is a consequence of higher-order terms in the Taylor expansion of $y(\theta)$ becoming significant at larger $|\varepsilon|$, reflecting the inherent nonlinearity of the circuit's input–output relationship.

Overall, the mitral response to the pinned average direction is modest but noticeably larger than when stepping along the sparse null-space directions from Section 4.9. This was expected since the average direction contains a nonzero projection onto eigenvectors of $\mathbf{H}$ with large eigenvalues due to the n_g component. For reference, Figure 34 also includes the response when stepping along three random directions, which produce substantially larger output changes.

The predicted parameter changes (Figure 33) reveal a coordinated pattern. In the case of the green cells, multiple changes act together to suppress their activity. The AIS shrinks, lowering intrinsic excitability (decreased n_g), blue cells send stronger inhibitory signals onto green cells (increased $w_{b,g}$), and excitatory drive onto both green and red cells is reduced (decreased $w_{osn,g}$ and $w_{r,g}$). Together, these changes suppress green cell activity in a consistent and coordinated manner.

According to the connectivity in Figure 10, maintaining mitral cell output under reduced green cell activity would be expected to involve compensatory changes such as reduced excitatory inputs ($w_{\text{osn},y}$ or $w_{r,y}$) and/or increased of other inhibitory inputs ($w_{b,y}$) onto the mitral cell. Interestingly, this is precisely what the analysis reveals, with small but detectable decreases in $w_{\text{osn},y}$ and increases in $w_{b,y}$ observed in Figure 33. These predictions remain to be experimentally validated, but offer concrete and testable targets for future work.

6 Conclusion

This work developed a biologically constrained computational model of the mouse olfactory bulb (OB) to investigate how the circuit maintains stable mitral cell output following structural changes induced by sensory deprivation. The central question, motivated by experimental observations from Galliano's lab, was to identify which parameter changes are consistent with the observed constant output of the OB circuit despite persistent reductions in axon initial segment (AIS) length in axonic (green) interneurons.

The analysis proceeded in two stages. In the first, we assumed the intrinsic excitability of all cell populations to remain constant during sensory deprivation, which in the model corresponds to holding all Hill coefficients n_j fixed. We characterised the output-preserving subspace of the resulting twelve-dimensional parameter space through a second-order Taylor expansion of the mitral cell output. A consistent four-dimensional near-null subspace of the Hessian was identified across all retained baseline solutions. Sparse rotation of the common subspace recovered four distinct families of interpretable basis vectors (Groups A–D), each identifying a specific combination of synaptic weights that can covary without altering circuit output. Comparison with experimental data revealed that one basis vector captures the observed increase in $w_{b,g}$, and further predicts compensatory increases in $w_{osn,g}$ and $w_{r,g}$, which are compatible with existing experimental observations.

However, these results are incomplete, as the Hill exponents n are assumed to be constant. To capture the experimentally observed axon initial segment (AIS) shrinkage, n_g is incorporated as a free parameter in the second stage of analysis. A second operating point was introduced to break the structural insensitivity of the circuit to n_g at $x = 1$, and a constrained direction analysis was used to identify compensatory adjustments in the remaining twelve parameters that minimise the resulting change in mitral cell output after pinning a change in n_g. The identified directions formed a cluster with similar covariation structure, and were also found to be approximately orthogonal to the output gradient. Stepping along the average direction produced output changes substantially smaller than those obtained from random perturbations of equivalent magnitude.

The model predicts a specific and coordinated set of changes: a decrease in $w_{\mathrm{osn},g}$, an increase in $w_{b,g}$, a decrease in $w_{r,g}$, and a reduction in n_g. All these changes contribute to the suppression of axonic (green) interneurons, in agreement with the experimental observed AIS shrinkage and the reduction in input–output relationship of axonic (green) cells. Since green cells are inhibitory onto mitral cells, preserving mitral cell output under reduced green cell activity requires compensatory adjustments. In particular, the model predicts an increase in inhibitory input from anaxonic (blue) cells ($w_{b,y}$) and a decrease in excitatory input from OSN sensory cells ($w_{\mathrm{osn},y}$). While these changes are consistent with existing experimental evidence and provide concrete targets for future investigation, the changes remain not entirely satisfactory, as perturbations in these directions produce small but measurable variations in mitral activity y.

Several natural extensions follow from this study. A first step would be to relax the assumption that changes in intrinsic excitability occur only in axonic green interneurons. Such changes are a general feature of homeostatic plasticity, and alterations in excitability have also been observed in other populations, such as ETC (red) cells.[8] More generally, the current model treats only the Hill coefficient as a variable measure of excitability. The Naka–Rushton functions also contain the semi-saturation constant b and maximum firing rate $y_{\max}$, both of which are currently implicit due to the model normalisation and become absorbed into the synaptic weight parameters w. Representing these quantities

explicitly and allowing them to vary would provide a more accurate description of changes in excitability and may alter the compensatory directions identified here.

Additionally, recent experimental work suggests that the glutamate release machinery in OSN terminals is upregulated following sensory deprivation[19] (Appendix B(B)). However, in the current model, the OSN input–output relationship is fixed. Introducing a Naka–Rushton function to describe OSN population activity would allow the model to account for experimentally observed OSN glutamate release increase.

A further extension would be to incorporate granule cells explicitly into the circuit. These are the most abundant interneuron population in the olfactory bulb and they provide inhibitory feedback onto mitral cells, which may play a more significant role in shaping circuit output than the current model assumes, e.g. by compensating for the observed silencing of inhibitory axonic (green) interneurons.

AI statement

Cursor, ChatGPT and Claude were used. All AI-generated text and code were carefully reviewed and edited. While most of the code was produced with AI assistance, the initial logic and overall direction were developed manually. AI tools were also used for debugging, code completion, plotting, and verifying mathematical derivations and understanding.

A Risk assessment statement

The primary risks associated with this project were related to prolonged computer use and extended periods of sedentary work. These included eye strain, neck and back discomfort, repetitive strain injuries, and fatigue associated with long durations of screen time and poor posture.

These risks were identified early on and mitigated against by maintaining regular exercise and movement breaks, improving workstation ergonomics by raising the screen to eye level and adjusting chair height to maintain a neutral arm and wrist position, and taking regular visual breaks by looking out of the window to reduce eye strain.

B Unpublished experimental data

The following data were provided by Dr E. Galliano via personal communication.

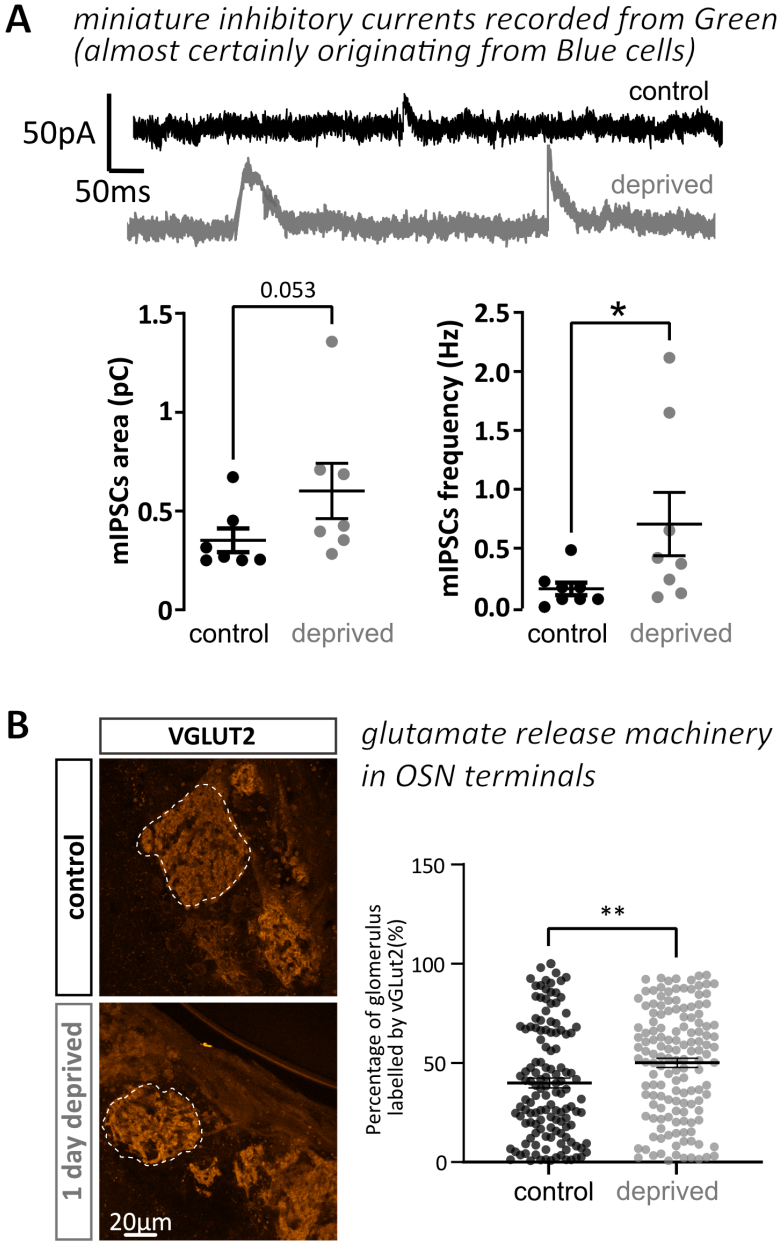


Figure 35

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